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Inventor(s)	Endo; Akio et al.

Flexible component support and display device

Abstract

A support for supporting a flexible component is provided. A flexible component support can be folded without decreasing the reliability of the flexible component. The support has a region where two different hinges overlap each other, and bending operation in the region can be performed only in one direction from a flat surface state. Accordingly, even in the case where unintended bending stress is applied to the region in an opposite direction, the flexible component or the support can be protected. In addition, the two hinges can each include position correction mechanism, which enables stable bending operation.

Inventors:	**Endo; Akio (Kanagawa, JP), Kusumoto; Naoto (Kanagawa, JP)**
Applicant:	**SEMICONDUCTOR ENERGY LABORATORY CO., LTD.** (Atsugi, JP)
Family ID:	**1000008777597**
Assignee:	**Semiconductor Energy Laboratory Co., Ltd. (Kanagawa-ken, JP)**
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7027110	12/2005	Akiyama et al.	N/A	N/A
7032984	12/2005	Kim et al.	N/A	N/A
7311366	12/2006	Kim et al.	N/A	N/A
8369075	12/2012	Huang	N/A	N/A
8837126	12/2013	Cho et al.	N/A	N/A
8988405	12/2014	Endo	N/A	N/A
9337434	12/2015	Lindblad	N/A	N/A
9395070	12/2015	Endo	N/A	N/A
9810406	12/2016	Endo	N/A	N/A
9876059	12/2017	Yanagisawa et al.	N/A	N/A
9891725	12/2017	Lindblad	N/A	N/A
9923156	12/2017	Jeong	N/A	N/A
10034399	12/2017	Shin et al.	N/A	N/A
10101826	12/2017	Lindblad	N/A	N/A
10119683	12/2017	Hirakata et al.	N/A	N/A
10367043	12/2018	Yanagisawa et al.	N/A	N/A
10578284	12/2019	Endo	N/A	N/A
10606379	12/2019	Lindblad	N/A	N/A
10727435	12/2019	Kim et al.	N/A	N/A
10736224	12/2019	Park et al.	N/A	N/A
10823374	12/2019	Endo	N/A	N/A
11063094	12/2020	Yanagisawa et al.	N/A	N/A
11800747	12/2022	Yanagisawa et al.	N/A	N/A
2002/0104769	12/2001	Kim et al.	N/A	N/A
2004/0183958	12/2003	Akiyama et al.	N/A	N/A
2006/0244693	12/2005	Yamaguchi et al.	N/A	N/A
2007/0097014	12/2006	Solomon et al.	N/A	N/A
2008/0055831	12/2007	Satoh	N/A	N/A
2009/0289877	12/2008	Kwon et al.	N/A	N/A
2010/0066643	12/2009	King et al.	N/A	N/A
2010/0308335	12/2009	Kim et al.	N/A	N/A

2011/0050657	12/2010	Yamada	N/A	N/A
2011/0286157	12/2010	Ma	N/A	N/A
2012/0033354	12/2011	Huang	N/A	N/A
2012/0307423	12/2011	Bohn et al.	N/A	N/A
2013/0010405	12/2012	Rothkopf et al.	N/A	N/A
2013/0083496	12/2012	Franklin et al.	N/A	N/A
2013/0187839	12/2012	Yang et al.	N/A	N/A
2013/0342090	12/2012	Ahn et al.	N/A	N/A
2014/0028596	12/2013	Seo et al.	N/A	N/A
2014/0065326	12/2013	Lee et al.	N/A	N/A
2014/0111954	12/2013	Lee et al.	N/A	N/A
2014/0159002	12/2013	Lee et al.	N/A	N/A
2014/0159020	12/2013	Song et al.	N/A	N/A
2014/0217382	12/2013	Kwon et al.	N/A	N/A
2014/0217875	12/2013	Park et al.	N/A	N/A
2014/0226275	12/2013	Ko et al.	N/A	N/A
2014/0240985	12/2013	Kim et al.	N/A	N/A
2014/0300529	12/2013	Kim et al.	N/A	N/A
2014/0307396	12/2013	Lee	N/A	N/A
2015/0016126	12/2014	Hirakata et al.	N/A	N/A
2015/0023031	12/2014	Endo	N/A	N/A
2015/0198978	12/2014	Catchpole	N/A	N/A
2015/0248149	12/2014	Yamazaki et al.	N/A	N/A
2015/0261259	12/2014	Endo et al.	N/A	N/A
2015/0277496	12/2014	Reeves et al.	N/A	N/A
2015/0325804	12/2014	Lindblad	N/A	N/A
2016/0007441	12/2015	Matsueda	N/A	N/A
2016/0118616	12/2015	Hiroki et al.	N/A	N/A
2016/0165718	12/2015	Oh	N/A	N/A
2016/0172623	12/2015	Lee	N/A	N/A
2016/0273739	12/2015	Endo	N/A	N/A
2016/0313811	12/2015	Lindblad	N/A	N/A
2016/0358967	12/2015	Madurawe et al.	N/A	N/A
2017/0142847	12/2016	Park	N/A	N/A
2018/0058665	12/2017	Endo	N/A	N/A
2018/0242466	12/2017	Lee	N/A	H05K 5/0017
2019/0018458	12/2018	Turchin et al.	N/A	N/A
2019/0036068	12/2018	Kim et al.	N/A	N/A
2019/0082544	12/2018	Park	N/A	N/A
2020/0004357	12/2019	Harada et al.	N/A	N/A
2020/0043382	12/2019	Isa et al.	N/A	N/A
2020/0192495	12/2019	Lindblad	N/A	N/A
2020/0321551	12/2019	Kim et al.	N/A	N/A
2020/0365827	12/2019	Yamazaki et al.	N/A	N/A
2021/0165454	12/2020	Dong	N/A	G06F 1/1652
2023/0350563	12/2022	Yamazaki et al.	N/A	N/A
2023/0384879	12/2022	Lindblad	N/A	N/A
2024/0094775	12/2023	Hirakata et al.	N/A	N/A

2024/0255789	12/2023	Hirakata	N/A	N/A
2024/0337366	12/2023	Hirakata et al.	N/A	N/A
2024/0357750	12/2023	Hirakata et al.	N/A	N/A
2024/0402860	12/2023	Matsumoto	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
001383503	12/2001	CN	N/A
106463079	12/2016	CN	N/A
106486517	12/2016	CN	N/A
106537485	12/2016	CN	N/A
107067976	12/2016	CN	N/A
108369440	12/2017	CN	N/A
109272872	12/2018	CN	N/A
208431261	12/2018	CN	N/A
109308850	12/2018	CN	N/A
109872638	12/2018	CN	N/A
202010008706	12/2010	DE	N/A
102018117701	12/2018	DE	N/A
1307805	12/2002	EP	N/A
3136714	12/2016	EP	N/A
3382497	12/2017	EP	N/A
2567044	12/2018	GB	N/A
2004-507779	12/2003	JP	N/A
3164599	12/2009	JP	N/A
2015-038609	12/2014	JP	N/A
2015-130320	12/2014	JP	N/A
2015-228022	12/2014	JP	N/A
2018-189968	12/2017	JP	N/A
2019-028467	12/2018	JP	N/A
2002-0012881	12/2001	KR	N/A
2002-0029698	12/2001	KR	N/A
2002-0034664	12/2001	KR	N/A
2002-0050062	12/2001	KR	N/A
2015-0010640	12/2014	KR	N/A
2017-0005019	12/2016	KR	N/A
2017-0057500	12/2016	KR	N/A
10-1834793	12/2017	KR	N/A
2018-0076271	12/2017	KR	N/A
2018-0097195	12/2017	KR	N/A
2019-0111163	12/2018	KR	N/A
522775	12/2002	TW	N/A
M395340	12/2009	TW	N/A
1610158	12/2017	TW	N/A
201911614	12/2018	TW	N/A
WO-2002/017051	12/2001	WO	N/A
WO-2014/057241	12/2013	WO	N/A
WO-2015/170213	12/2014	WO	N/A
WO-2016/012900	12/2015	WO	N/A

OTHER PUBLICATIONS

International Search Report (Application No. PCT/IB2020/053247) Dated Jul. 28, 2020 (7 pages). cited by applicant
Written Opinion (Application No. PCT/IB2020/053247) Dated Jul. 28, 2020 (8 pages). cited by applicant
Taiwanese Office Action (Application No. 109111111) Dated Dec. 26, 2023. cited by applicant

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Primary Examiner: Hossain; Moazzam

Attorney, Agent or Firm: NIXON PEABODY LLP

Background/Summary

TECHNICAL FIELD
(1) The present invention relates to an object, a method, or a manufacturing method. Alternatively, the present invention relates to a process, a machine, manufacture, or a composition of matter. In particular, one embodiment of the present invention relates to a semiconductor device, a light-emitting device, a display device, an electronic device, a lighting device, a driving method thereof, or a manufacturing method thereof. In particular, one embodiment of the present invention relates to a flexible component support and a display device including the flexible component support.
(2) Note that in this specification and the like, a semiconductor device generally means a device that can function by utilizing semiconductor characteristics. A transistor, a semiconductor circuit, an arithmetic device, a memory device, and the like are each one embodiment of the semiconductor device. Moreover, a light-emitting device, a display device, a lighting device, and an electronic device each include a semiconductor device in some cases.
BACKGROUND ART
(3) Electronic devices such as cellular phones, smartphones, tablet type computers, and laptop computers are fabricated with appropriate size depending on their functions, usability, and portability. Meanwhile, it is inconvenient to carry a plurality of electronic devices.
(4) Accordingly, a mode in which a plurality of electronic devices can be integrated is desirable. For example, Patent Document 1 discloses a tri-fold light-emitting panel. The use of the light-emitting panel enables integration of functions of a plurality of electronic devices and manufacture of an electronic device whose size is variable.
PRIOR ART DOCUMENT
Patent Document
(5) [Patent Document 1] Japanese Published Patent Application No. 2015-130320
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
(6) The portability of a display panel formed over a flexible substrate can be improved when the display panel is folded. On the other hand, when the display panel is bent in a direction opposite to design intention, for example, the display panel, housings, a hinge portion for connecting the housings, or the like might be broken; therefore, the display panel is preferably well designed so as not to be easily bent in the opposite direction.
(7) Thus, an object of one embodiment of the present invention is to provide a support for supporting a flexible component. Another object is to provide a support for performing bending operation without decreasing the reliability of a flexible component. Another object is to provide a novel flexible component support. Another object is to provide a novel light-emitting device.
(8) Another object is to provide a folding display device with high portability. Another object is to

provide a folding display device with high display visibility. Another object is to provide a folding display device having a power saving function. Another object is to provide a novel display device. Another object is to provide an operation method of a novel display device.
(9) Note that the description of these objects does not preclude the existence of other objects. One embodiment of the present invention does not need to achieve all these objects. In addition, objects other than the above will be apparent from the description of the specification and the like, and objects other than the above can be derived from the description of the specification and the like.
Means for Solving the Problems
(10) One embodiment of the present invention relates to a support of a flexible display panel or a tri-fold folding display device with high portability.
(11) One embodiment of the present invention is a flexible component support that includes a first housing, a second housing, a first joint, and a second joint. The first housing and the second housing are coupled to each other through the first joint and the second joint. The first joint and the second joint have an overlap region. The first joint and the second joint each include a movable portion. The first joint includes a plurality of first columnar bodies. The plurality of first columnar bodies are coupled to each other so that first surfaces of the plurality of first columnar bodies form a continuous surface. The second joint is a hinge. The maximum opening angle of the hinge is approximately 180°.
(12) A second columnar body including a major axis substantially perpendicular to the first surface is fixed to one of the plurality of first columnar bodies. The hinge includes a notch portion. When the opening angle of the hinge is approximately 180°, part of the second columnar body can be positioned in the notch portion.
(13) Alternatively, one of end portions in a major axis direction of a second columnar body including a major axis substantially perpendicular to the first surface may be fixed to one of the plurality of first columnar bodies. The hinge may include a shaft that is not fixed to a blade. The other of the end portions in the major axis direction of the second columnar body may be fixed to the shaft.
(14) In addition, another embodiment of the present invention is a flexible component support that includes a first housing, a second housing, a first joint, and a second joint. The first housing and the second housing each include a first surface and a second surface positioned opposite to the first surface. The first housing and the second housing are coupled to each other through the first joint and the second joint. The first joint and the second joint have a region where the first joint and the second joint overlap each other. The first joint and the second joint each include a movable portion. The first joint includes a plurality of first columnar bodies. A cross section perpendicular to a major axis of the first columnar body is a substantial trapezoid. The first columnar body includes a first side surface including one of legs of the substantial trapezoid, a second side surface including the other of the legs of the substantial trapezoid, and a third side surface including a lower side of the substantial trapezoid. The two adjacent first columnar bodies have a structure in which the first side surface of one of the first columnar bodies is adjacent to a second side surface of the other of the first columnar bodies and the third side surfaces of the first columnar bodies are coupled to each other to form a continuous surface. The second joint is a hinge that includes a first blade and a second blade. The maximum angle formed by the first blade and the second blade is approximately 180°. The first joint and the second joint are capable of shifting the first surfaces of the first housing and the second housing from a state where the first surfaces of the first housing and the second housing face in the same direction to a state where the first surfaces of the first housing and the second housing face each other.
(15) A structure can be employed in which the third side surface of the first columnar body is continuous with the second surface of the first housing and the second surface of the second housing.
(16) The number of the first columnar bodies can be an odd number. A second columnar body

including a major axis substantially perpendicular to the third side surface of the first columnar body can be fixed to the first columnar body positioned in a center. Each of the first blade and the second blade can be provided with a notch portion. When the angle formed by the first blade and the second blade is approximately 180°, part of the second columnar body can be positioned in the notch portion.
(17) Two connection parts can be provided between the first blade and the second blade, and the notch portion can be provided between the two connection parts.
(18) Alternatively, the number of the first columnar bodies may be an odd number. One of end portions in a major axis direction of a second columnar body including a major axis substantially perpendicular to the third side surface of the first columnar body may be fixed to the first columnar body positioned in a center. The hinge may include a first shaft fixed to neither the first blade nor the second blade. The other of the end portions in the major axis direction of the second columnar body may be fixed to the first shaft.
(19) A second shaft can be fixed to the first blade, a third shaft can be fixed to the second blade, and the first shaft can be provided between the second shaft and the third shaft.
(20) The first blade includes a region overlapping the first surface of the first housing. The first blade and the first surface of the first housing are capable of sliding each other. The second blade includes a region overlapping the first surface of the second housing. The second blade and the first surface of the second housing are capable of sliding each other.
(21) In addition, a structure may be employed in which the flexible component support further includes a third housing and a third joint. The third housing may include a first surface and a second surface positioned on a side opposite to that of the first surface. The second housing and the third housing may be coupled to each other through the third joint. The third joint may include a movable portion. The third joint may include a plurality of third columnar bodies. A cross section perpendicular to a major axis of the third columnar body may be a substantial rectangle. The third columnar body may include a fourth side surface including one side of the substantial rectangle, a fifth side surface facing the fourth side surface, and a sixth side surface perpendicular to the fourth side surface and the fifth side surface. The two adjacent third columnar bodies may have a structure in which the fourth side surface of one of the third columnar bodies is adjacent to the fifth side surface of the other of the third columnar bodies and the sixth side surfaces of the third columnar bodies are coupled to each other to form a continuous surface. The third joint may be capable of shifting the first surfaces of the second housing and the third housing from a state where the first surfaces of the second housing and the third housing face in the same direction to a state where the second surfaces of the second housing and the third housing face each other.
(22) A structure can be employed in which the sixth side surface of the third columnar body is continuous with the second surface of the second housing and the second surface of the third housing.
(23) A display device can be constructed when a flexible display panel is provided in the flexible component support.
(24) In the flexible component support including the first to third housings, a structure can be employed in which a flexible display panel is provided over and across the second surface of the first housing to the second surface of the third housing to construct a display device.
(25) It is preferable that the display panel include a light-emitting device.
Effect of the Invention
(26) With the use of one embodiment of the present invention, it is possible to provide a support for supporting a flexible component. Alternatively, it is possible to provide a support for performing bending operation without decreasing the reliability of a flexible component. Alternatively, it is possible to provide a novel flexible component support. Alternatively, it is possible to provide a novel light-emitting device.
(27) Alternatively, it is possible to provide a folding display device with high portability.

Alternatively, it is possible to provide a folding display device with high display visibility. Alternatively, it is possible to provide a folding display device having a power saving function. Alternatively, it is possible to provide a folding display device that is easy to hold. Alternatively, it is possible to provide a novel display device. Alternatively, it is possible to provide an operation method of a display device.

(28) Note that the description of these effects does not preclude the existence of other effects. Note that one embodiment of the present invention does not need to have all of these effects. Note that effects other than these will be apparent from the description of the specification, the drawings, the claims, and the like and effects other than these can be derived from the description of the specification, the drawings, the claims, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A and FIG. **1**B are diagrams illustrating a support.
(2) FIG. **2**A and FIG. **2**B are diagrams illustrating the support.
(3) FIG. **3**A and FIG. **3**B are diagrams illustrating the support.
(4) FIG. **4**A to FIG. **4**C are diagrams illustrating the support.
(5) FIG. **5**A and FIG. **5**B are diagrams illustrating a support.
(6) FIG. **6**A to FIG. **6**C are diagrams illustrating notch mechanism.
(7) FIG. **7**A and FIG. **7**B are diagrams illustrating a support.
(8) FIG. **8**A and FIG. **8**B are diagrams illustrating the support.
(9) FIG. **9**A and FIG. **9**B are diagrams illustrating the support.
(10) FIG. **10**A to FIG. **10**C are diagrams illustrating the support.
(11) FIG. **11**A to FIG. **11**C are diagrams illustrating a display device.
(12) FIG. **12**A to FIG. **12**C are diagrams illustrating a hinge portion.
(13) FIG. **13**A to FIG. **13**C are diagrams illustrating the hinge portion.
(14) FIG. **14**A to FIG. **14**C are diagrams each illustrating operation of a display device.
(15) FIG. **15**A to FIG. **15**C are diagrams each illustrating operation of a display device.
(16) FIG. **16**A to FIG. **16**C are diagrams each illustrating operation of a display device.
(17) FIG. **17**A and FIG. **17**B are diagrams each illustrating an application example of a display device.
(18) FIG. **18**A to FIG. **18**D are diagrams each illustrating an application example of the display device.
(19) FIG. **19**A and FIG. **19**B are diagrams each illustrating an application example of a display device.
(20) FIG. **20** is a diagram illustrating a structure example of a display panel.
(21) FIG. **21** is a diagram illustrating a structure example of the display panel.
(22) FIG. **22** is a diagram illustrating a structure example of a display panel.
(23) FIG. **23**A is a block diagram of a display panel. FIG. **23**B and FIG. **23**C are pixel circuit diagrams.
(24) FIG. **24**A, FIG. **24**C, and FIG. **24**D are pixel circuit diagrams. FIG. **24**B is a timing chart showing pixel operation.
(25) FIG. **25**A, FIG. **25**B, FIG. **25**C, FIG. **25**D, and FIG. **25**E are diagrams each illustrating a pixel structure example.

MODE FOR CARRYING OUT THE INVENTION

(26) Embodiments will be described in detail with reference to the drawings. Note that the present invention is not limited to the following description, and it will be readily understood by those skilled in the art that modes and details of the present invention can be modified in various ways

without departing from the spirit and scope of the present invention. Therefore, the present invention should not be construed as being limited to the description of embodiments below. Note that in structures of the invention described below, the same reference numerals are used in common, in different drawings, for the same portions or portions having similar functions, and a repeated description thereof is omitted in some cases. Note that the hatching of the same component that constitutes a drawing is sometimes omitted or changed as appropriate in different drawings.

(27) In addition, even in the case where a single component is illustrated in a circuit diagram, the component may be composed of a plurality of parts as long as there is no functional inconvenience. For example, in some cases, a plurality of transistors that operate as a switch are connected in series or in parallel. Furthermore, in some cases, capacitors are divided and arranged in a plurality of positions.

(28) In addition, one conductor has a plurality of functions such as a wiring, an electrode, and a terminal in some cases. In this specification, a plurality of names are used for the same component in some cases. Furthermore, even in the case where elements are illustrated in a circuit diagram as if they were directly connected to each other, the elements may actually be connected to each other through one conductor or a plurality of conductors. In this specification, even such a structure is included in the category of direct connection.

Embodiment 1

(29) In this embodiment, a flexible component support and a display device according to one embodiment of the present invention are described with reference to drawings. Note that in this specification, a display panel is typically treated as a flexible component; however, the flexible component may be another component. Examples of the flexible component include a solar battery, a primary battery, a secondary battery, an antenna, a speaker, a microphone, a cable, illumination, a variety of terminals, a variety of sensors, a variety of circuits, a composite device including any of these, and the like.

(30) In addition, in this specification, a display device generally means a device that has a display function. In other words, an electronic device including a display portion is included in a display device. For example, an electronic device including a display portion, such as a cellular phone, a smartphone, a smartwatch, a tablet type computer, or a television, is included in a display device.

(31) One embodiment of the present invention is a flexible component support capable of folding a flexible component. The support has a region where bending operation can be performed only in one direction from a flat surface state. Accordingly, even in the case where unintended bending stress is applied to the region in an opposite direction, the flexible component or the support can be protected.

(32) In addition, another embodiment of the present invention is a foldable display device including a flexible display panel. The display device has tri-fold mechanism, and a region where folding is performed so that first surfaces of the display device face each other and a region where folding is performed so that surfaces opposite to the first surfaces face each other can be formed. Accordingly, even a display panel with a comparatively high aspect ratio of a display surface, for example, 16:9, 18:9, or 21:9 can be folded small when a crease is provided in a minor axis direction, which results in improved portability. Furthermore, a display region that cannot be seen when being folded is made to display no images, so that power consumption can be significantly reduced.

(33) <Flexible Component Support **1**>

(34) FIG. **1**A is a diagram illustrating a state where a flexible component support **100**A according to one embodiment of the present invention is unfolded. FIG. **1**B is a diagram illustrating a cross section cut along a surface F**1** perpendicular to the support **100**A. In addition, a plane view, a front view, and a side view that correspond to the state in FIG. **1**A are illustrated in FIG. **4**A, FIG. **4**B, and FIG. **4**C, respectively.

(35) The support **100**A includes a housing **102***a*, a housing **102***b*, a hinge **10**, and a hinge **20**. The hinge **10** and the hinge **20** have a region where the hinge **10** and the hinge **20** overlap each other in the same movable direction.
(36) Note that a housing refers to a box-like member whose inside or surface can be provided with a component or the like. In the case where a component is provided for only a surface of a housing, the housing may be a plate-like member. Alternatively, the entire part including a hinge is sometimes referred to as a housing. In addition, a hinge refers to a joint including a movable portion and is a component that controls the relative position of two members coupled to each other. Furthermore, a housing and a hinge are sometimes formed continuously using the same member.
(37) A housing in this embodiment is a substantially rectangular solid that includes a first surface, a second surface opposite to the first surface, and other surfaces. In addition, although the first surface and the second surface are illustrated as flat surfaces in this embodiment, the first surface and the second surface may include curved surfaces. Furthermore, the surface may be provided with step height or the like, as needed. Alternatively, a corner of a rectangular solid may be rounded. Moreover, one of the first surface and the second surface can be referred to as an upper surface, and the other of the first surface and the second surface can be referred to as a lower surface, for example.
(38) The hinge **10** has a hinge structure and includes a blade **11***a*, a blade **11***b*, a shaft **12***a*, a shaft **13***a*, a shaft **12***b*, a shaft **13***b*, a spindle **14**, a spindle **15**, and a plurality of stoppers **17**.
(39) The shaft **12***a* and the shaft **13***a* are fixed to the blade **11***a*. The shaft **12***b* and the shaft **13***b* are fixed to the blade **11***b*. The spindle **14** is inserted into the shaft **12***a* and the shaft **12***b* so that a first connection part is formed. The spindle **15** is inserted into the shaft **13***a* and the shaft **13***b* so that a second connection part is formed. With the above structure, the blade **11***a* and the blade **11***b* can move with the first and second connection parts used as rotation axes (fulcrums).
(40) Note that the shaft **12***a* and the shaft **13***a* may be parts of the blade **11***a*. In addition, the shaft **12***b* and the shaft **13***b* may be parts of the blade **11***b*.
(41) In addition, a notch portion **30** is provided for the blade **11***a* and the blade **11***b* between the first connection part and the second connection part (see FIG. **4**A). The notch portion **30** is used for position correction together with a pin **16** to be described later.
(42) The blade **11***a* has a region overlapping a first surface of the housing **102***a*. In addition, the blade **11***a* has a region overlapping part of a jig **19** fixed to the housing **102***a*. Gaps are provided between the blade **11***a* and the housing **102***a* and between the blade **11***a* and the jig **19**. Accordingly, the blade **11***a* and the first surface of the housing **102***a* can slide over each other.
(43) The blade **11***b* has a region overlapping a first surface of the housing **102***b*. In addition, the blade **11***b* has a region overlapping part of the jig **19** fixed to the housing **102***b*. Gaps are provided between the blade **11***b* and the housing **102***b* and between the blade **11***b* and the jig **19**. Accordingly, the blade **11***b* and the first surface of the housing **102***b* can slide over each other.
(44) Note that the stoppers **17** are provided over the blade **11***a* and the blade **11***b*, and contact of the jigs **19** with the stoppers **17** can limit the slide amount. In addition, providing the stoppers **17** can prevent separation of the hinge **10**.
(45) Note that slide mechanism may be constructed with a structure different from that in FIG. **1**A. For example, as illustrated in FIG. **5**A, a structure may be employed in which slotted holes **22** are provided in the blade **11***a* and stoppers **23** are provided through the slotted holes **22**.
(46) The stopper **23** includes a small axis whose diameter is smaller than a minor axis of the slotted hole **22** and a top part whose diameter is larger than the minor axis of the slotted hole **22**. The axis is fixed to the housing **102***a* through the slotted hole **22**. Gaps are provided between the housing **102***a* and the blade **11***a* and between the blade **11***a* and the top part of the stopper **23**. Accordingly, the blade **11***a* and the first surface of the housing **102***a* can slide over each other. A blade **11***b* side can also have a similar structure. Moreover, slide mechanism using a rail or a bearing may be

employed.
(47) As illustrated in FIG. **4**A, a region **31** where an end surface of the blade **11***a* is in contact with an end surface of the blade **11***b* is generated in the hinge **10** in an unfolded state. Here, when an angle formed by a top surface (a side on which a spindle is provided) of each blade and the end surface is approximately 90°, the maximum opening angle (angle formed by the blade **11***a* and the blade **11***b*) of the hinge **10** can be approximately 180°. Accordingly, the hinge **10** is compatible with bending in one direction but is not compatible with bending in an opposite direction. In other words, even when the hinge **20** is compatible with bending in the opposite direction, the hinge **10** can inhibit bending in the opposite direction; therefore, even when bending stress is applied to the flexible component or the support in the opposite direction, the flexible component or the support can be protected.
(48) Alternatively, as illustrated in FIG. **5**B, a backing plate **24** for preventing reverse bending may be provided on a blade lower surface. When the backing plate **24** that is fixed to one of the blades is in contact with the other of the blades, further bending cannot be performed. Accordingly, the backing plate **24** is made in contact with the blade when the angle formed by the blade **11***a* and the blade **11***b* is approximately 180°, so that the maximum opening angle of the hinge **10** can be controlled so as to be approximately 180°.
(49) In that case, the angle formed by the top surface and the end surface of the blade does not necessarily have adequate accuracy. Note that although FIG. **5**B illustrates an example where the backing plate **24** is fixed to the blade **11***b*, the backing plate **24** may be fixed to the blade **11***a*. In addition, the backing plate **24** may be part of the blade **11***a* (the blade **11***b*).
(50) The hinge **20** includes a plurality of columnar bodies **21** each with a cross section perpendicular to a major axis used as a trapezoid or a substantial trapezoid. The columnar body **21** includes a first side surface (a surface including one of legs of the trapezoid or the substantial trapezoid) and a second side surface (a surface including the other of the legs of the trapezoid or the substantial trapezoid). Two adjacent columnar bodies **21** have a structure where the first side surface of one of the columnar bodies **21** is adjacent to the second side surface of the other of the columnar bodies **21**. Some of the two adjacent columnar bodies **21** are directly connected to each other or connected to each other through another member, and their relative positions can be changed.
(51) The columnar bodies **21** are coupled to each other such that third side surfaces (surfaces each including a lower base of the trapezoid or the substantial trapezoid) form a continuous surface. In addition, the third side surface of the columnar body **21** in one of end portions of the hinge **20** is connected to the second surface of the housing **102***a* so as to be continuous with the second surface of the housing **102***a*. Furthermore, the third side surface of the columnar body **21** in the other of the end portions of the hinge **20** is connected to the second surface of the housing **102***b* so as to be continuous with the second surface of the housing **102***b*. Note that the shape of a fourth side surface (a surface including an upper base of the trapezoid or the substantial trapezoid) of each of the columnar bodies **21** may be any shape as long as no interference occurs in the other columnar body **21**, the hinge **10**, and the housings **102***a* and **102***b*. Accordingly, the cross section perpendicular to the major axis of the columnar body **21** may be a triangle or a substantial triangle.
(52) Note that a substantial trapezoid refers to a shape that can be substantially regarded as a trapezoid. For example, examples of the substantial trapezoid include a shape that includes curves in some sides of a trapezoid, a shape in which a trapezoid corner is rounded, and the like. The same applies to a substantial triangle.
(53) The number of the columnar bodies **21** included in the hinge **20** is an odd number. The pin **16** is fixed to the columnar body **21** positioned in the center, and a major axis of the pin **16** and the third side surface of the columnar body **21** form a substantially perpendicular angle. The pin **16** is a columnar body, and part of the pin **16** is positioned in the notch portion **30** when the support **100**A is in an unfolded state, as illustrated in FIG. **1**B. Note that the shape of a cross section

perpendicular to the major axis of the pin **16** is a circle, for example. In that case, a top surface shape of the notch portion **30** included in one blade of the hinge **10** is preferably a semicircle that is larger than a curvature radius of the pin **16**.
(54) FIG. **2**A is a diagram illustrating a state where bending operation is added to the support **100**A. In addition, FIG. **2**B is a diagram illustrating a cross section cut along the surface F**1** perpendicular to the support **100**A.
(55) As illustrated in FIG. **2**A and FIG. **2**B, when bending operation is added to the support **100**A, the first side surface of the one of the two adjacent columnar bodies **21** and the second side surface of the other of the columnar bodies **21** in the hinge **20** are transformed to be close to each other. In that case, the third side surfaces of the two adjacent columnar bodies **21** are continuous with each other to form a certain angle, and a region where the entire cross section is substantially arc-shaped is formed. Accordingly, when the flexible component is provided to overlap the region, the component can form a curved surface in a portion overlapping the region.
(56) When the cross section of the hinge **20** is transformed to be arc-shaped, the length of an inner arc in the hinge **20** is smaller than the length of an outer arc. Accordingly, the blade **11***a* of the hinge **10** positioned inside the hinge **20** slides to offset the change. The same applies to the blade **11***b*.
(57) The hinge **10**, and the housing **102***a* and the housing **102***b* form simple slide mechanism. Thus, one of the blade **11***a* and the blade **11***b* sometimes slides more widely than the other of the blade **11***a* and the blade **11***b*. When the one of the blade **11***a* and the blade **11***b* slides too widely, troubles such as deviation of a bending center position and inability of bending operation afterwards sometimes occur.
(58) In one embodiment of the present invention, an end portion of the notch portion **30** in the one of the blades that slides widely is in contact with the pin **16** so that slide operation is forcibly stopped and slide operation of the other of the blades is promoted. Accordingly, a rotation axis of the hinge **10** can be corrected to be positioned around the center of the hinge **20**, and stable bending operation can be performed.
(59) FIG. **3**A is a diagram illustrating a folded state where bending operation is further added to the support **100**A. In addition, FIG. **3**B is a diagram illustrating a cross section cut along the surface F**1** perpendicular to the support **100**A. Note that in FIG. **3**A and FIG. **3**B, the housing **102***a* is illustrated by a dashed line for clarity.
(60) As illustrated in FIG. **3**B, even when the support **100**A is in a completely folded state, neither the hinge **10** nor the hinge **20** interferes with the pin **16**.
(61) In addition, also in the course of changing the support **100**A from the folded state illustrated in FIG. **3**A to the unfolded state illustrated in FIG. **1**A through the state in FIG. **2**A, the one of the blade **11***a* and the blade **11***b* in the hinge **10** sometimes slides more widely than the other of the blade **11***a* and the blade **11***b* in the hinge **10**.
(62) As in the above, the end portion of the notch portion **30** in the one of the blades that slides widely is in contact with the pin **16** so that slide operation is forcibly stopped and slide operation of the other of the blades is promoted also in this case. Accordingly, the rotation axis of the hinge **10** can be corrected to be positioned around the center of the hinge **20**, and stable unfold operation can be performed.
(63) Note that although the above description is made under a concept that the position of the hinge **10** is corrected with respect to the operation of the hinge **20**, it can also be said that the position of the hinge **20** is relatively corrected with respect to the operation of the hinge **10**. Alternatively, it can also be said that the hinge **10** and the hinge **20** also correct their positions each other.
(64) In addition, as illustrated in FIG. **6**A, the hinge **10** may be provided with notch mechanism. The notch mechanism can include a lock part **29** including a ball **28** on a tip of a spring **27**, a hole portion **25** provided in the shaft **12***b*, and hollow portions **26** provided in the shaft **12***a*. Note that although the notch mechanism provided for the shaft **12***a* and the shaft **12***b* is described here,

similar notch mechanism may be provided for the shaft **13***a* and the shaft **13***b*.
(65) When a state is set in which the lock part **29** is inserted into the hole portion **25** and the shaft **12***a* and the shaft **12***b* are combined with each other, as illustrated in an inside view of FIG. **6**B, the ball **28** enters the hollow portion **26** due to elasticity of the spring **27** and is simply locked. When bending operation is performed with equal to or more than a certain amount of force, the ball **28** is unlocked and the blade is moved, and the ball **28** enters another hollow portion **26** and is simply locked again.
(66) For example, in the case where five hollow portions **26** are provided at intervals of 45° starting from 180° with the spindle **14** as the center, the ball **28** can be simply locked when an angle formed by the two blades is 180°, 135°, 90°, 45°, or 0°. When the ball **28** is locked at an appropriate angle, unintended transformation when using or carrying can be prevented. Alternatively, user's convenience can be improved when using at the same angle every time, for example.
(67) Note that in the hollow portion **26** with a shallow shape as illustrated in FIG. **6**B, the ball **28** is unlocked comparatively easily, which enables rotation in both directions for increasing and decreasing the angle. In contrast, in the hollow portion **26** with a deep asymmetrical shape as illustrated in FIG. **6**C, rotation in a direction for increasing the angle can be inhibited, for example.
(68) A slope is formed in one region of the hollow portion **26** illustrated in FIG. **6**C, and the ball **28** can therefore move. However, there is no slope in a region opposite to the one region, and the ball **28** cannot move. Accordingly, for example, when the hollow portion **26** in FIG. **6**C is used as the hollow portion **26** in which the angle formed by the two blades is approximately 180°, the maximum angle formed by the blade **11***a* and the blade **11***b* can be controlled so as to be approximately 180°.
(69) <Flexible Component Support **2**>
(70) FIG. **7**A is a diagram illustrating a state where a support **100**B that is different from the support **100**A is unfolded. The structure of the hinge **10** in the support **100**B is different from that in the support **100**A. FIG. **7**B is a diagram illustrating a cross section cut along the surface F**1** perpendicular to the support **100**B is a cut surface. In addition, a plane view, a front view, and a side view that correspond to the state in FIG. **7**A are illustrated in FIG. **10**A, FIG. **10**B, and FIG. **10**C, respectively.
(71) In the following description, the hinge **10** included in the support **100**B is mainly described, and a description of the housings **102***a* and **102***b* and the hinge **20** that are common to the support **100**A and a description of a slide between a blade of the hinge **10** and a housing are omitted. In addition, components common to the support **100**A are denoted by the same reference numerals.
(72) The hinge **10** included in the support **100**B has a hinge structure and includes the blade **11***a*, the blade **11***b*, a shaft **41***a*, a shaft **41***b*, a shaft **42**, a spindle **45**, and the plurality of stoppers **17**.
(73) The shaft **41***a* is fixed to the blade **11***a*. The shaft **41***b* is fixed to the blade **11***b*. The shaft **42** is fixed to neither the blades **11***a* nor **11***b*. The shaft **42** is provided between the shaft **41***a* and the shaft **41***b*, and the spindle **45** is inserted into the shaft **41***a*, the shaft **42**, and the shaft **41***b* so that a connection part is formed. With the above structure, the blade **11***a* and the blade **11***b* can move with the connection part used as a rotation axis (fulcrum).
(74) Note that the shaft **41***a* may be part of the blade **11***a*. In addition, the shaft **41***b* may be part of the blade **11***b*. A pin **46** is fixed to the shaft **42**, and the pin **46** is fixed to one of the columnar bodies **21** included in the hinge **20**.
(75) As illustrated in FIG. **10**A and FIG. **10**B, the region **31** where the end surface of the blade **11***a* is in contact with the end surface of the blade **11***b* is generated in the hinge **10** in an unfolded state. Accordingly, like the hinge **10** included in the support **100**A, the maximum opening angle of the hinge **10** can be approximately 180°, and bending in an opposite direction can be inhibited.
(76) Like the support **100**A, the number of the columnar bodies **21** included in the hinge **20** is an odd number, and one of end portions in a major axis direction of the pin **46** where the third side surface of the columnar body **21** and the major axis form a substantially perpendicular angle is

fixed to the columnar body **21** positioned in the center, as illustrated in FIG. **7**B and FIG. **10**B, for example. The pin **46** is a columnar body, and the other of the end portions in the major axis direction of the pin **46** is fixed to the shaft **42**. Note that the number of the pins **46** is not limited to one and may be plural. In addition, the shape of a cross section perpendicular to the major axis of the pin **46** is not particularly limited.
(77) FIG. **8**A is a diagram illustrating a state where bending operation is added to the support **100**B. In addition, FIG. **8**B is a diagram illustrating a cross section cut along the surface F**1** perpendicular to the support **100**B.
(78) As illustrated in FIG. **8**A and FIG. **8**B, when bending operation is added to the support **100**B, the third side surfaces of the columnar bodies **21** included in the hinge **20** are continuous with each other to form a certain angle; thus, a substantially arc-shaped region is formed in a cross section of the entire hinge **20**. In that case, the length of an inner arc in the hinge **20** is smaller than the length of an outer arc. Accordingly, the blade **11***a* of the hinge **10** positioned inside the hinge **20** slides to offset the change. The same applies to the blade **11***b*.
(79) In the support **100**B, the shaft **42** that is positioned in the center of the rotation axis (connection part) of the hinge **10** is fixed to the columnar body **21** that is positioned in the center of the hinge **20** through the pin **46**; thus, the rotation axis of the hinge **10** and a central part of the hinge **20** operate so as to always overlap each other. Accordingly, one of the blade **11***a* and the blade **11***b* does not slide more widely than the other of the blade **11***a* and the blade **11***b*, so that stable bending operation can be performed.
(80) FIG. **9**A is a diagram illustrating a folded state where bending operation is further added to the support **100**B. In addition, FIG. **9**B is a diagram illustrating a cross section cut along the surface F**1** perpendicular to the support **100**B. Note that in FIG. **9**A and FIG. **9**B, the housing **102***a* is illustrated by a dashed line for clarity.
(81) Also in the course of changing the support **100**B from the folded state illustrated in FIG. **9**A to the unfolded state illustrated in FIG. **7**A through the state in FIG. **8**A, stable unfold operation can be performed because the rotation axis of the hinge **10** and the central part of the hinge **20** operate so as to always overlap each other.
(82) Note that the structures of the slotted hole **22** of the hinge **10** and the stopper **23** that are illustrated in FIG. **5**A can also be employed in the support **100**B. In addition, the structure of the backing plate **24** illustrated in FIG. **5**B can also be employed. Furthermore, the structure of the notch mechanism illustrated in FIG. **6**A and FIG. **6**B can also be employed.
(83) <Display Device>
(84) The support **100**A or the support **100**B that is provided with a flexible display panel can be employed in a display device.
(85) FIG. **11**A to FIG. **11**C are diagrams illustrating a tri-fold display device. Note that a combination of the hinge **10** and the hinge **20** that are included in the support **100**A is described here as a hinge portion **101***a*. The hinge portion **101***a* is provided between the housing **102***a* and the housing **102***b*. A hinge portion **101***b* is provided between the housing **102***b* and a housing **102***c*. Here, like the other housings, the housing **102***c* is a substantially rectangular solid with a structure including a first surface and a second surface. A flexible display panel **103** can be provided over the second surfaces of the housings **102***a* to **102***c* that are opposite to the first surfaces of the housings **102***a* to **102***c*.
(86) FIG. **11**A is a perspective view on the first surface sides of the housings **102***a* to **102***c* with a display device in an unfolded state. Although a structure where two hinges **10** are provided for the hinge portion **101***a* is illustrated, one hinge **10** may be provided as in FIG. **1**A and the like. Alternatively, three or more hinges **10** may be provided. The number of the hinges **10** may be determined in consideration of the housing width so that stable bending operation can be performed.
(87) A structure illustrated in FIG. **12**A to FIG. **12**C can be used for the hinge portion **101***b*, for

example.
(88) The hinge portion **101***b* includes a plurality of columnar bodies **115** each with a rectangular or substantially rectangular cross section perpendicular to a major axis. The columnar body **115** includes a first side surface (a surface including one side of the rectangle or the substantial rectangle) and a second side surface facing the first side surface. Two adjacent columnar bodies **115** have a structure where the first side surface of one of the columnar bodies **115** is adjacent to the second side surface of the other of the columnar bodies **115**. Some of the two adjacent columnar bodies **115** are directly connected to each other or connected to each other through another member, and their relative positions can be changed.
(89) The columnar bodies **115** are coupled to each other such that third side surfaces that are perpendicular to the first side surfaces and the second side surfaces form a continuous surface. In addition, the third side surface of the columnar body **115** in one of end portions of the hinge portion **101***b* is connected to the second surface of the housing **102***b* so as to be continuous with the second surface of the housing **102***b*. Furthermore, the third side surface of the columnar body **115** in the other of the end portions of the hinge portion **101***b* is connected to the second surface of the housing **102***c* so as to be continuous with the second surface of the housing **102***c*. Note that the shape of a fourth side surface facing the third side surface of the columnar body **115** may be any shape as long as no interference occurs in the other columnar body.
(90) As illustrated in FIG. **12**A, when the first side surface of the one of the two adjacent columnar bodies **115** and the second side surface of the other of the columnar bodies **115** are transformed in a direction in which they are apart from each other, the flexible display panel can be set in a folded state. In that case, the third side surfaces of the two adjacent columnar bodies **115** are continuous with each other to form a certain angle, and a substantially arc-shaped region is formed in a cross section of the entire hinge portion **101***b*. Accordingly, in the flexible display panel, the component can form a concave curved surface in a portion overlapping the region.
(91) When transformation operation (unfold operation) is performed from the state in FIG. **12**A, the two adjacent columnar bodies **115** move in a direction where the first side surface of the one of the columnar bodies **115** becomes close to the second side surface of the other of the columnar bodies **115**, as illustrated in FIG. **12**B, and a curvature radius of the substantial arc changes so as to be larger. In that case, a curvature radius of a curved surface portion changes so as to be larger also in the display panel.
(92) When transformation operation is further performed from the state in FIG. **12**B, as illustrated in FIG. **12**C, the second surface of the housing **102***b*, the third side surface of each of the columnar bodies **115**, and the second surface of the housing **102***c* are continuous with each other so as to be flat. In that case, the curved surface portion of the display panel also changes so as to be flat, and the entire display panel is set in an unfolded state so as to be flat. When transformation operation is performed in an order opposite to the above, the display panel can be folded.
(93) Note that since the cross section of the columnar body **115** is a rectangle, when the display panel is unfolded so as to be flat, in the two adjacent columnar bodies **115**, the first side surface of the one of the columnar bodies **115** is in contact with the second side surface of the other of the columnar bodies **115**. Accordingly, the hinge portion **101***b* does not cause bending of the display panel in an opposite direction, and mechanism to inhibit reverse bending may be unnecessary. Note that a spacer for keeping a gap between the housings during folding may be provided. In addition, the housing or the hinge may be transformed into a shape suitable for installation of the display panel as appropriate.
(94) FIG. **13**A to FIG. **13**C are diagrams illustrating another example of the hinge portion **101***b*.
(95) The hinge portion **101***b* includes a gear **116***a* and a gear **116***b*. The gear **116***a* is fixed to the housing **102***b*. The gear **116***b* is fixed to the housing **102***c*. A central axis of the gear **116***a* preferably overlaps the second surface of the housing **102***b* that is opposite to the first surface of the housing **102***b*. In addition, a central axis of the gear **116***b* preferably overlaps the second surface

of the housing **102***c* that is opposite to the first surface of the housing **102***c*.
(96) As illustrated in FIG. **13**A, the gear **116***a* and the gear **116***b* are engaged with each other at a certain position in a folded state. In that case, the central axes of the two gears are on the second surfaces of the housings; thus, a gap is generated between the housings (between display surfaces of the display panel that face each other). Therefore, the flexible display panel can form a curved surface whose curvature radius is about the half of the gap.
(97) When transformation operation (unfold operation) is performed from the state in FIG. **13**A, the housing **102***b* and the housing **102***c* are in synchronization with each other in accordance with engagement between the gear **116***a* and the gear **116***b* and move to be opened with the hinge portion **101***b* used as a fulcrum (see FIG. **13**B). In that case, a curvature radius of a curved surface portion changes so as to be larger also in the display panel.
(98) When transformation operation is further performed from the state in FIG. **13**B, as illustrated in FIG. **13**C, the second surface of the housing **102***b* and the first surface of the housing **102***c* are continuous with each other so as to be flat. In that case, the curved surface portion of the display panel also changes so as to be flat, and the entire display panel is set in an unfolded state so as to be flat. When transformation operation is performed in an order opposite to the above, the display panel can be folded.
(99) Note that mechanism for keeping the engagement between the gear **116***a* and the gear **116***b* may be provided. In addition, when the display panel is unfolded so as to be flat, a side surface of the housing **102***b* is in contact with a side surface of the housing **102***c*. Accordingly, the hinge portion **101***b* does not cause bending of the display panel in an opposite direction, and mechanism to inhibit reverse bending may be unnecessary. Note that a spacer for keeping a gap between the housings during folding may be provided. Alternatively, mechanism for keeping the gap may be provided in the gear **116***a* and the gear **116***b*. Furthermore, the housing or the hinge may be transformed into a shape suitable for installation of the display panel as appropriate.
(100) FIG. **11**B is a perspective view on the second surface sides of the housings **102***a* to **102***c* with a display device in an unfolded state. The flexible display panel **103** is provided on the second surface sides of the housings **102***a* to **102***c*.
(101) Note that in this embodiment, for clarity of the description, the display panel **103** is divided into three regions: a region **103***a*, a region **103***b*, and a region **103***c*. The region **103***a*, the region **103***b*, and the region **103***c* are regions that are in parallel with a horizontal direction (a direction in which a surface of the display panel **103** extends) when the display panel **103** is unfolded so as to be flat and are regions bordered by a position where the hinge portion is provided or its vicinity. Note that actually, the regions **103***a* to **103***c* and their boundaries have no structural differences. One seamless flexible display panel can be used as the display panel **103**.
(102) FIG. **11**C is a perspective view illustrating a folded state of a display device. The hinge portion **101***a* is compatible with outward bending where the display surface of the display panel **103** is a convex, and the hinge portion **101***b* is compatible with inward bending where the display surface of the display panel **103** is a concave. Accordingly, the display panel **103** can be folded in three, as illustrated in FIG. **11**C.
Display Operation Example 1
(103) FIG. **14**A to FIG. **14**C are diagrams each illustrating an operation example of a display device according to one embodiment of the present invention. FIG. **14**A is operation of setting a curved surface **104***a* (part of the region **103***a* and part of the region **103***b*) in a non-display state when the display device is in a folded state and a flat portion of the region **103***a* is in a display state. In that case, as illustrated in a B**1**-B**2** cross section illustrated in FIG. **14**B, folded regions that cannot be seen (the region **103***b* and the region **103***c*) are also preferably set in a non-display state.
(104) Alternatively, as illustrated in FIG. **14**C, when the flat portion of the region **103***a* is in a non-display state, the curved surface **104***a* may be set in a display state. As in FIG. **14**A and FIG. **14**B, the folded regions that cannot be seen are also preferably set in a non-display state. In this manner,

when the display device is in a folded state and only some of the regions are set in a display state, power-saving operation can be performed.

Display Operation Example 2

(105) FIG. **15**A to FIG. **15**C are diagrams each illustrating an example in which a display portion of a display device according to one embodiment of the present invention is divided into three surfaces and utilized.

(106) FIG. **15**A is a diagram illustrating an example in which an angle formed by the housing **102***c* and the housing **102***b* is an obtuse angle and an angle formed by the housing **102***b* and the housing **102***a* is an acute angle so that the display device is set on a desk or the like in a balanced way. When the housing **102***a* is used as a leg, the display device can be utilized like a laptop computer. For example, operation can be performed by displaying a keyboard **131**, an icon **132**, and an image **130** of application software on the region **103***c*, a curved surface **104***b* (part of the region **103***b* and part of the region **103***c*), and the region **103***b*, respectively, and touching a screen.

(107) In that case, when a mode is employed in which the same image **130** is also displayed on the region **103***a*, as illustrated in FIG. **15**B, people on the opposite side can see the same image with high visibility. Alternatively, as illustrated in FIG. **15**C, the display device may operate in a power-saving mode while the region **103***a* is in a non-display state.

Display Operation Example 3

(108) FIG. **16**A to FIG. **16**C are diagrams each illustrating an example in which a display portion of a display device according to one embodiment of the present invention is divided into two surfaces and utilized.

(109) FIG. **16**A is a diagram illustrating an example in which an angle formed by the housing **102***a* and the housing **102***b* is approximately greater than or equal to 60° and less than 180° (for example, approximately 90° or the like) and an angle formed by the housing **102***b* and the housing **102***c* is approximately 180° so that the display device is set on a desk or the like in a balanced way. When a larger screen is made by forming a continuous plane of the region **103***b* and the region **103***c* and a display surface (the region **103***b* and the region **103***c*) is tilted with the housing **102***a* used as a leg, visibility can be increased.

(110) In that case, as illustrated in FIG. **16**B, the display device may operate in a power-saving mode while the region **103***a* is in a non-display state.

(111) FIG. **16**C is a diagram illustrating an example in which an angle formed by the housing **102***c* and the housing **102***b* is approximately less than 180° and greater than or equal to 90° (for example, approximately 135° or the like) and an angle formed by the housing **102***b* and the housing **102***a* is approximately 180° so that the display device is set on a desk or the like in a balanced way. When the housing **102***a* and the housing **102***b* are placed in parallel with a plane, input using a stylus **150** or the like can be facilitated. In addition, when the region **103***c* is tilted, the visibility can be increased.

Application Example 1

(112) FIG. **17**A and FIG. **17**B are diagrams each illustrating an example in which a display device illustrated in this embodiment is applied to an information terminal such as a smartphone. Note that components common to the display device are denoted by the same reference numerals. A display device **200** includes audio input/output units **135***a* and **135***b*, cameras **136***a* and **136***b*, a sensor **137**, and a sensor **120**.

(113) When one of the audio input/output units **135***a* and **135***b* functions as a microphone, the other of the audio input/output units **135***a* and **135***b* can function as a speaker. Therefore, when a telephone function is utilized, for example, it is possible to have a conversation without any inconvenience even when the audio input/output units **135***a* and **135***b* are held in either direction. The microphone function and the speaker function can be switched with each other by the sensor **120** that senses a tilt. In addition, either the cameras **136***a* and **136***b* can similarly function with a high priority by the sensor **120**.

(114) The input/output units **135***a* and **135***b* may include both of a device that functions as a microphone and a device that functions as a speaker or may include one device that has both functions.
(115) Alternatively, both the input/output units **135***a* and **135***b* can function as microphones to record stereo sound. Alternatively, both the input/output units **135***a* and **135***b* can function as speakers to reproduce stereo sound.
(116) In addition, both the cameras **136***a* and **136***b* can function and can capture 3D images. The sensor **137** is an optical sensor and can adjust display luminance so that the images can be easily seen in accordance with ambient illuminance.
(117) In addition, as illustrated in FIG. **17**B, a display panel **138** may be provided on a rear surface that is on a side opposite to a front surface of the display device **200** where the display panel **103** is provided. The display panel **138** can display the same image as the display panel **103**. The display panel **138** can also be utilized as a sub-display that displays simple information, painting, a pattern, a picture, or the like; lighting; or the like. A display panel using a light-emitting device or a liquid crystal device can be used as the display panel **138**, and low-power electronic paper or the like may be used as the display panel **138**. A display panel using a hard substrate as a support can also be used as the display panel **138**.
(118) Note that as illustrated in FIG. **18**A, the display panel **138** may be provided in each of the housings **102***a* to **102***c*. Alternatively, as illustrated in FIG. **18**B, a flexible display panel **139** may be provided on the rear surface of the display device **200**. In that case, the display panel **139** can be bent, so that the display panel **139** can be provided across the housings **102***a* to **102***c*, like the display panel **103** provided on the front surface.
(119) Alternatively, as illustrated in FIG. **18**C, a solar battery **140** may be provided on the rear surface of the display device **200**. A battery in the display device **200** can be charged with power generated by the solar battery **140**, and the power can be supplied to the outside through an external interface **145**.
(120) Note that FIG. **18**C illustrates an example of a solar battery including a hard support. As the solar battery, for example, a silicon solar battery using crystalline silicon for a photoelectric conversion layer, a solar battery with a tandem structure of a silicon solar battery and a perovskite type solar battery, or the like can be used.
(121) Alternatively, as illustrated in FIG. **18**D, a solar battery using a flexible substrate as a support may be provided on the rear surface of the display device **200**. As the solar battery, for example, a thin-film solar battery **141** such as an amorphous silicon solar battery, a CIGS (Cu—In—Ga—Se) type solar battery, an organic solar battery, or a perovskite type solar battery, or the like can be used. The solar battery using a flexible substrate as a support can be provided across the housings **102***a* to **102***c*, like the display panel **139**.
Application Example 2
(122) FIG. **19**A and FIG. **19**B are diagrams each illustrating an example in which a display portion of a display device according to one embodiment of the present invention is used properly depending on the application.
(123) FIG. **19**A and FIG. **19**B are diagrams each illustrating an example in which a display device illustrated in this embodiment is applied to an order terminal in a restaurant or the like. Note that components common to the display device are denoted by the same reference numerals.
(124) A display device **210** includes a transceiving unit **146**, a speaker **147**, a camera **148**, a microphone **149**, and the like. Note that the display device **210** may have functions of a general tablet type computer in addition to functions of one embodiment of the present invention.
(125) In normal times, the display device **210** can be set in a folded state as illustrated in FIG. **19**A, and a function of calling over a salesclerk or an intercom function can be utilized. When the display device **210** is unfolded, a menu can be displayed and an order can be placed. Ordered items can be transmitted through the transceiving unit **146**. In addition, display of the total amount of order and

payment with a barcode captured by the camera **148** can be performed.
(126) Note that as the shape of the display device **210** when being unfolded, it is preferable that the housing **102***a* and the housing **102***b* form an obtuse angle and that the housing **102***b* and the housing **102***c* form an angle of approximately 90°. With such a shape, the housing **102***c* can be used as a leg and can be folded easily.
(127) At least part of the structure examples, the drawings corresponding thereto, and the like exemplified in this embodiment can be implemented in combination with the other structure examples, the other drawings, and the like as appropriate.
(128) At least part of this embodiment can be implemented in combination with the other embodiments described in this specification as appropriate.
Embodiment 2
(129) In this embodiment, a structure example of a display panel that can be applied to the display device according to one embodiment of the present invention is described.
Structure Example
(130) FIG. **20** illustrates a top view of a display panel **700**. The display panel **700** is a display panel that employs a flexible support substrate **745** and can be used as a flexible display. In addition, the display panel **700** includes a pixel portion **702** provided over the flexible support substrate **745**. Furthermore, a source driver circuit portion **704**, a pair of gate driver circuit portions **706**, a wiring **710**, and the like are provided over the support substrate **745**. Moreover, a plurality of display devices are provided in the pixel portion **702**.
(131) In addition, part of the support substrate **745** is provided with an FPC terminal portion **708** to which an FPC **716** (FPC: Flexible printed circuit) is connected. The pixel portion **702**, the source driver circuit portion **704**, and the gate driver circuit portions **706** are each supplied with a variety of signals and the like from the FPC **716** through the FPC terminal portion **708** and the wiring **710**.
(132) The pair of gate driver circuit portions **706** is provided on both sides with the pixel portion **702** sandwiched therebetween. Note that the gate driver circuit portions **706** and the source driver circuit portion **704** may be formed separately on semiconductor substrates or the like to form packaged IC chips. The IC chips can be mounted on the support substrate **745** by a COF (Chip On Film) technique or the like.
(133) Transistors including an OS are preferably applied to transistors included in the pixel portion **702**, the source driver circuit portion **704**, and the gate driver circuit portions **706**.
(134) Light-emitting devices or the like can be used as the display devices in the pixel portion **702**. Examples of light-emitting devices are self-luminous light-emitting devices such as an LED (Light Emitting Diode), an OLED (Organic LED), a QLED (Quantum-dot LED), a semiconductor laser, and the like. Alternatively, liquid crystal devices such as transmissive liquid crystal devices, reflective liquid crystal devices, or semi-transmissive liquid crystal devices can also be used as the display devices. Alternatively, shutter type or optical interference type MEMS (Micro Electro Mechanical Systems) devices; display devices employing a microcapsule method, an electrophoretic method, an electrowetting method, an Electronic Liquid Powder (registered trademark) method, or the like; and the like can also be used.
(135) In addition, FIG. **20** illustrates an example where the support substrate **745** has a shape in which a portion provided with the FPC terminal portion **708** is projected. In a region P**1** in FIG. **20**, part of the support substrate **745** that includes the FPC terminal portion **708** can be bent backward. Bending the part of the support substrate **745** backward enables the FPC **716** to be placed in a state overlapping a rear side of the pixel portion **702** when the display panel **700** is mounted on an electronic device or the like, so that the electronic device or the like can be space-saving or small-sized.
(136) In addition, an IC **717** is mounted on the FPC **716** connected to the display panel **700**. The IC **717** has a function of a source driver circuit, for example. In that case, a structure can be employed in which the source driver circuit portion **704** in the display panel **700** includes at least one of a

protection circuit, a buffer circuit, a demultiplexer circuit, and the like.
Cross-Sectional Structure Example
(137) Structures using organic EL as the display device are described below using FIG. **21** and FIG. **22**. Each of FIG. **21** and FIG. **22** is a schematic cross-sectional view of the display panel **700** illustrated in FIG. **20** along a dash-dot line S-T.
(138) First, portions common to the display panels illustrated in FIG. **21** and FIG. **22** are described.
(139) FIG. **21** and FIG. **22** illustrate cross sections including the pixel portion **702**, the gate driver circuit portion **706**, and the FPC terminal portion **708**. The pixel portion **702** includes a transistor **750** and a capacitor **790**. The gate driver circuit portion **706** includes a transistor **752**.
(140) Each of the transistor **750** and the transistor **752** is a transistor in which an oxide semiconductor is employed for a semiconductor layer where a channel is formed. Note that the transistors are not limited thereto, and a transistor using silicon (amorphous silicon, polycrystalline silicon, or single-crystal silicon) or a transistor using an organic semiconductor for a semiconductor layer can be employed.
(141) The transistor used in this embodiment includes a highly purified oxide semiconductor film in which formation of oxygen vacancies is suppressed. The transistor can have extremely low off-state current. Accordingly, in the pixel employing such a transistor, the retention time of an electric signal such as an image signal can be extended, and an interval between writings of an image signal or the like can also be set longer. Accordingly, the frequency of refresh operations can be reduced, so that power consumption can be reduced.
(142) In addition, the transistor used in this embodiment can have comparatively high field-effect mobility and thus is capable of high-speed driving. For example, with such a transistor capable of high-speed driving used for the display panel, a switching transistor in a pixel portion and a driver transistor used in a driver circuit portion can be formed over the same substrate. In other words, a structure in which a driver circuit formed using a silicon wafer or the like is not employed is possible, and the number of components of the display device can be reduced. Furthermore, the use of the transistor capable of high-speed driving also in the pixel portion can provide a high-quality image.
(143) The capacitor **790** includes a lower electrode formed by processing the same film as a film for a first gate electrode of the transistor **750** and an upper electrode formed by processing the same metal oxide film as a film for the semiconductor layer. The upper electrode has reduced resistance like a source region and a drain region of the transistor **750**. In addition, part of an insulating film functioning as a first gate insulating layer of the transistor **750** is provided between the lower electrode and the upper electrode. That is, the capacitor **790** has a stacked-layer structure in which an insulating film functioning as a dielectric film is sandwiched between a pair of electrodes. Furthermore, a wiring obtained by processing the same film as a film for a source electrode and a drain electrode of the transistor **750** is connected to the upper electrode.
(144) In addition, an insulating layer **770** that functions as a planarization film is provided over the transistor **750**, the transistor **752**, and the capacitor **790**.
(145) As the transistor **750** included in the pixel portion **702** and the transistor **752** included in the source driver circuit portion **704**, transistors having different structures may be used. For example, a structure may be employed in which a top-gate transistor is used as one of the transistors **750** and **752** and a bottom-gate transistor is used as the other of the transistors **750** and **752**. Note that the same applies to the source driver circuit portion **704**, as in the gate driver circuit portion **706**.
(146) The FPC terminal portion **708** includes a wiring **760** part of which functions as a connection electrode, an anisotropic conductive film **780**, and the FPC **716**. The wiring **760** is electrically connected to a terminal included in the FPC **716** through the anisotropic conductive film **780**. Here, the wiring **760** is formed using the same conductive film as the source electrode and the drain electrode of the transistor **750** or the like.
(147) Next, the display panel **700** illustrated in FIG. **21** is described.

(148) The display panel **700** illustrated in FIG. **21** includes the support substrate **745** and a support substrate **740**. As each of the support substrate **745** and the support substrate **740**, a glass substrate or a flexible substrate such as a plastic substrate can be used, for example.
(149) The transistor **750**, the transistor **752**, the capacitor **790**, and the like are provided over the insulating layer **744**. The support substrate **745** and the insulating layer **744** are attached to each other with an adhesive layer **742**.
(150) In addition, the display panel **700** includes a light-emitting device **782**, a coloring layer **736**, a light-blocking layer **738**, and the like.
(151) The light-emitting device **782** includes a conductive layer **772**, an EL layer **786**, and a conductive layer **788**. The conductive layer **772** is electrically connected to the source electrode or the drain electrode included in the transistor **750**. The conductive layer **772** is provided over the insulating layer **770** and functions as a pixel electrode. In addition, an insulating layer **730** is provided to cover an end portion of the conductive layer **772**. Over the insulating layer **730** and the conductive layer **772**, the EL layer **786** and the conductive layer **788** are stacked and provided.
(152) For the conductive layer **772**, a material having a property of reflecting visible light can be used. For example, a material including aluminum, silver, or the like can be used. For the conductive layer **788**, a material having a property of transmitting visible light can be used. For example, an oxide material including indium, zinc, tin, or the like is preferably used. Thus, the light-emitting device **782** is a top-emission light-emitting device that emits light to a side opposite a formation surface (a support substrate **740** side).
(153) The EL layer **786** includes an organic compound or an inorganic compound such as quantum dots. The EL layer **786** includes a light-emitting material that exhibits blue light when current flows.
(154) As the light-emitting material, a fluorescent material, a phosphorescent material, a thermally activated delayed fluorescence (TADF) material, an inorganic compound (a quantum dot material or the like), or the like can be used. Examples of materials that can be used for quantum dots include a colloidal quantum dot material, an alloyed quantum dot material, a core-shell quantum dot material, a core quantum dot material, and the like.
(155) The light-blocking layer **738** and the coloring layer **736** are provided on one surface of an insulating layer **746**. The coloring layer **736** is provided in a position overlapping the light-emitting device **782**. In addition, the light-blocking layer **738** is provided in a region not overlapping the light-emitting device **782** in the pixel portion **702**. Furthermore, the light-blocking layer **738** may also be provided to overlap the gate driver circuit portion **706** or the like.
(156) The support substrate **740** is attached to the other surface of the insulating layer **746** with an adhesive layer **747**. In addition, the support substrate **740** and the support substrate **745** are attached to each other with a sealing layer **732**.
(157) Here, for the EL layer **786** included in the light-emitting device **782**, a light-emitting material that exhibits white light emission is employed. White light emission by the light-emitting device **782** is colored by the coloring layer **736** to be emitted to the outside. The EL layer **786** is provided over the pixels that exhibit different colors. The pixels provided with the coloring layer **736** transmitting any of red (R), green (G), and blue (B) are arranged in a matrix in the pixel portion, so that the display panel **700** can perform full-color display.
(158) In addition, a conductive film having a semi-transmissive property and a semi-reflective property may be used for the conductive layer **788**. In that case, a microcavity structure is achieved between the conductive layer **772** and the conductive layer **788** such that light of a specific wavelength can be intensified to be emitted. Also at this time, a structure may be employed in which an optical adjustment layer for adjusting an optical distance is placed between the conductive layer **772** and the conductive layer **788** such that the thickness of the optical adjustment layer differs between pixels of different colors and accordingly the color purity of light emitted from each pixel is increased.

(159) Note that a structure in which the coloring layer **736** or the optical adjustment layer is not provided may be employed when the EL layer **786** is formed into an island shape for each pixel or into a stripe shape for each pixel column, i.e., the EL layer **786** is formed by separate coloring.
(160) Here, an inorganic insulating film that functions as a barrier film having low permeability is preferably used for each of the insulating layer **744** and the insulating layer **746**. With a structure in which the light-emitting device **782**, the transistor **750**, and the like are sandwiched between the insulating layer **744** and the insulating layer **746**, degradation thereof can be inhibited and a highly reliable display panel can be achieved.
(161) In a display panel **700**A illustrated in FIG. **22**, a resin layer **743** is provided between the adhesive layer **742** and the insulating layer **744** illustrated in FIG. **21**. In addition, a protective layer **749** is provided instead of the support substrate **740**.
(162) The resin layer **743** is a layer including an organic resin such as polyimide or acrylic. The insulating layer **744** includes an inorganic insulating film of silicon oxide, silicon oxynitride, silicon nitride, or the like. The resin layer **743** and the support substrate **745** are attached to each other with the adhesive layer **742**. The resin layer **743** is preferably thinner than the support substrate **745**.
(163) The protective layer **749** is attached to the sealing layer **732**. A glass substrate, a resin film, or the like can be used as the protective layer **749**. Alternatively, as the protective layer **749**, an optical member such as a polarizing plate (including a circularly polarizing plate) or a scattering plate, an input device such as a touch sensor panel, or a structure in which two or more of these are stacked may be employed.
(164) In addition, the EL layer **786** included in the light-emitting device **782** is provided over the insulating layer **730** and the conductive layer **772** in an island shape. The EL layers **786** are formed separately so that respective subpixels emit light of different colors, so that color display can be performed without the use of the coloring layer **736**.
(165) In addition, a protective layer **741** is provided to cover the light-emitting device **782**. The protective layer **741** has a function of preventing diffusion of impurities such as water into the light-emitting device **782**. The protective layer **741** has a stacked-layer structure in which an insulating layer **741***a*, an insulating layer **741***b*, and an insulating layer **741***c* are stacked in this order from the conductive layer **788** side. In that case, it is preferable that inorganic insulating films with a high barrier property against impurities such as water be used as the insulating layer **741***a* and the insulating layer **741***c* and an organic insulating film that functions as a planarization film be used as the insulating layer **741***b*. Furthermore, the protective layer **741** is preferably provided to extend also to the gate driver circuit portion **706**.
(166) In addition, an organic insulating film covering the transistor **750**, the transistor **752**, and the like is preferably formed in an island shape inward from the sealing layer **732**. In other words, an end portion of the organic insulating film is preferably inward from the sealing layer **732** or in a region overlapping an end portion of the sealing layer **732**. FIG. **22** illustrates an example in which the insulating layer **770**, the insulating layer **730**, and the insulating layer **741***b* are processed into island shapes. The insulating layer **741***c* and the insulating layer **741***a* are provided in contact with each other in a portion overlapping the sealing layer **732**, for example. Thus, when a structure is employed in which a surface of the organic insulating film covering the transistor **750** and the transistor **752** is not exposed to the outside of the sealing layer **732**, diffusion of water or hydrogen from the outside to the transistor **750** and the transistor **752** through the organic insulating film can be favorably prevented. This can reduce variations in electrical characteristics of the transistors, so that a display device with extremely high reliability can be achieved.
(167) In addition, in FIG. **22**, the region P**1** that can be bent includes a portion where the support substrate **745**, the adhesive layer **742**, and the inorganic insulating film such as the insulating layer **744** are not provided. Furthermore, the region P**1** has a structure in which the insulating layer **770** including an organic material covers the wiring **760** so that the wiring **760** is not exposed. When a

structure is employed in which an inorganic insulating film is not provided as long as possible in the region P**1** that can be bent and only a conductive layer containing a metal or an alloy and a layer containing an organic material are stacked, generation of cracks caused at bending can be prevented. Moreover, when the support substrate **745** is not provided in the region P**1**, part of the display panel **700**A can be bent with an extremely small curvature radius.
(168) In addition, in FIG. **22**, a conductive layer **761** is provided over the protective layer **741**. The conductive layer **761** can be used as a wiring or an electrode.
(169) In addition, in the case where a touch sensor is provided to overlap the display panel **700**A, the conductive layer **761** can function as an electrostatic shielding film for preventing transmission of electrical noise to the touch sensor during pixel driving. In that case, a structure in which a predetermined constant potential is applied to the conductive layer **761** may be employed.
(170) Alternatively, the conductive layer **761** can be used as an electrode of the touch sensor, for example. This enables the display panel **700**A to function as a touch panel. For example, the conductive layer **761** can be used as an electrode or a wiring of a capacitive touch sensor. In that case, the conductive layer **761** can be used as a wiring or an electrode to which a sensor circuit is connected or a wiring or an electrode to which a sensor signal is input. When the touch sensor is formed over the light-emitting device **782** in this manner, the number of components can be reduced, and manufacturing cost of an electronic device or the like can be reduced.
(171) The conductive layer **761** is preferably provided in a portion not overlapping the light-emitting device **782**. The conductive layer **761** can be provided in a position overlapping the insulating layer **730**, for example. Thus, a transparent conductive film with a comparatively low conductivity is not necessarily used for the conductive layer **761**, and a metal or an alloy having high conductivity or the like can be used, so that the sensitivity of the sensor can be increased.
(172) Note that as the type of the touch sensor that can be formed using the conductive layer **761**, a variety of types such as a resistive type, a surface acoustic wave type, an infrared type, an optical type, and a pressure-sensitive type can be used, without limitation to a capacitive type. Alternatively, two or more of these may be combined and used.
(173) <Components>
(174) Components such as a transistor that can be employed in the display device will be described below.
(175) [Transistor]
(176) The transistors each include a conductive layer functioning as a gate electrode, a semiconductor layer, a conductive layer functioning as a source electrode, a conductive layer functioning as a drain electrode, and an insulating layer functioning as a gate insulating layer.
(177) Note that there is no particular limitation on the structure of the transistor included in the display device according to one embodiment of the present invention. For example, a planar transistor, a staggered transistor, or an inverted staggered transistor may be used. In addition, a top-gate or bottom-gate transistor structure may be employed. Alternatively, gate electrodes may be provided above and below a channel.
(178) There is no particular limitation on the crystallinity of a semiconductor material used for the transistors, and any of an amorphous semiconductor and a crystalline semiconductor (a microcrystalline semiconductor, a polycrystalline semiconductor, a single-crystal semiconductor, or a semiconductor partly including crystal regions) may be used. It is preferable that a crystalline semiconductor be used because degradation of the transistor characteristics can be suppressed.
(179) In particular, a transistor that uses a metal oxide film for a semiconductor layer where a channel is formed will be described below.
(180) As a semiconductor material used for the transistors, a metal oxide whose energy gap is greater than or equal to 2 eV, preferably greater than or equal to 2.5 eV, further preferably greater than or equal to 3 eV can be used. Typically, an oxide semiconductor containing indium, or the like can be used, and for example, a CAAC-OS, a CAC-OS, or the like described later can be used. A

CAAC-OS has a crystal structure including stable atoms and is suitable for a transistor that puts emphasis on reliability, and the like. A CAC-OS exhibits excellent mobility characteristics and thus is suitable for a transistor that is driven at high speed, for example.
(181) In an OS transistor, a semiconductor layer has a large energy gap, and thus the OS transistor can exhibit characteristics with an extremely low off-state current of several yoctoamperes per micrometer (a current value per micrometer of channel width). In addition, an OS transistor has features such that impact ionization, an avalanche breakdown, a short-channel effect, and the like do not occur, which are different from those of a Si transistor, and can form a highly reliable circuit. Furthermore, variations in electrical characteristics due to crystallinity unevenness, which are issues in Si transistors, are less likely to occur in OS transistors.
(182) A semiconductor layer can be, for example, a film represented by an In-M-Zn-based oxide that contains indium, zinc, and M (a metal such as aluminum, titanium, gallium, germanium, yttrium, zirconium, lanthanum, cerium, tin, neodymium, or hafnium). The In-M-Zn-based oxide can be formed by, for example, a sputtering method, an ALD (Atomic layer deposition) method, an MOCVD (Metal organic chemical vapor deposition) method, or the like.
(183) In the case where an In-M-Zn-based oxide is deposited by a sputtering method, it is preferable that the atomic ratio of metal elements in a sputtering target satisfy In≥M and Zn≥M. The atomic ratio of metal elements in such a sputtering target is preferably, for example, InM:Zn=1:1:1, In:M:Zn=1:1:1.2, In:M:Zn=3:1:2, In:M:Zn=4:2:3, In:M:Zn=4:2:4.1, In:M:Zn=5:1:6, In:M:Zn=5:1:7, In:M:Zn=5:1:8, or the like. Note that the atomic ratio in the deposited semiconductor layer varies from the atomic ratio of metal elements of the sputtering target in a range of ±40%.
(184) A metal oxide film with a low carrier density is used as the semiconductor layer. For example, for the semiconductor layer, a metal oxide whose carrier density is lower than or equal to 1×10.sup.17/cm.sup.3, preferably lower than or equal to 1×10.sup.15/cm.sup.3, further preferably lower than or equal to 1×10.sup.13/cm.sup.3, still further preferably lower than or equal to 1×10.sup.11/cm.sup.3, even further preferably lower than 1×10.sup.10/cm.sup.3, and higher than or equal to 1×10.sup.−9/cm.sup.3 can be used. Such a metal oxide is referred to as a highly purified intrinsic or substantially highly purified intrinsic metal oxide. The oxide semiconductor has a low density of defect states and thus can be regarded as a metal oxide having stable characteristics.
(185) Note that, without limitation to those described above, an oxide semiconductor with an appropriate composition may be used in accordance with required semiconductor characteristics and electrical characteristics (field-effect mobility, threshold voltage, and the like) of the transistor. In addition, to obtain the required semiconductor characteristics of the transistor, it is preferable that the carrier density, the impurity concentration, the density of defect states, the atomic ratio between a metal element and oxygen, the interatomic distance, the density, and the like of the semiconductor layer be set to appropriate values.
(186) When silicon or carbon, which is one of Group 14 elements, is contained in the metal oxide contained in the semiconductor layer, oxygen vacancies are increased in the semiconductor layer, and the semiconductor layer becomes n-type. Thus, the concentration of silicon or carbon (concentration obtained by secondary ion mass spectrometry) in the semiconductor layer is set to lower than or equal to 2×10.sup.18 atoms/cm.sup.3, preferably lower than or equal to 2×10.sup.17 atoms/cm.sup.3.
(187) In addition, alkali metal and alkaline earth metal might generate carriers when bonded to a metal oxide, in which case the off-state current of the transistor might be increased. Thus, the concentration of alkali metal or alkaline earth metal in the semiconductor layer that is obtained by secondary ion mass spectrometry is set to lower than or equal to 1×10.sup.18 atoms/cm.sup.3, preferably lower than or equal to 2×10.sup.16 atoms/cm.sup.3.
(188) In addition, when nitrogen is contained in the metal oxide contained in the semiconductor layer, electrons serving as carriers are generated and the carrier density increases, so that the

semiconductor layer easily becomes n-type. As a result, a transistor using a metal oxide that contains nitrogen is likely to have normally-on characteristics. Accordingly, the nitrogen concentration in the semiconductor layer that is obtained by secondary ion mass spectrometry is preferably set to lower than or equal to 5×10.sup.18 atoms/cm.sup.3.
(189) In addition, when hydrogen is contained in an oxide semiconductor included in the semiconductor layer, hydrogen reacts with oxygen bonded to a metal atom to be water, and thus sometimes causes an oxygen vacancy in the oxide semiconductor. When a channel formation region in the oxide semiconductor includes oxygen vacancies, the transistor sometimes has normally-on characteristics. Furthermore, in some cases, a defect that is an oxygen vacancy into which hydrogen enters functions as a donor and generates an electron serving as a carrier. In other cases, bonding of part of hydrogen to oxygen bonded to a metal atom generates electrons serving as carriers. Thus, a transistor using an oxide semiconductor that contains a large amount of hydrogen is likely to have normally-on characteristics.
(190) A defect in which hydrogen has entered an oxygen vacancy can function as a donor of the oxide semiconductor. However, it is difficult to evaluate the defect quantitatively. Thus, the oxide semiconductor is sometimes evaluated by not its donor concentration but its carrier concentration. Therefore, in this specification and the like, the carrier concentration assuming the state where an electric field is not applied is sometimes used, instead of the donor concentration, as the parameter of the oxide semiconductor. That is, “carrier concentration” described in this specification and the like can be replaced with “donor concentration” in some cases.
(191) Therefore, hydrogen in the oxide semiconductor is preferably reduced as much as possible. Specifically, the hydrogen concentration in the oxide semiconductor that is obtained by secondary ion mass spectrometry (SIMS) is set to lower than 1×10.sup.20 atoms/cm.sup.3, preferably lower than 1×10.sup.19 atoms/cm.sup.3, further preferably lower than 5×10.sup.18 atoms/cm.sup.3, still further preferably lower than 1×10.sup.18 atoms/cm.sup.3. When an oxide semiconductor with a sufficiently low concentration of impurities such as hydrogen is used for a channel formation region of a transistor, the transistor can have stable electrical characteristics.
(192) In addition, oxide semiconductors (metal oxides) can be classified into a single crystal oxide semiconductor and a non-single-crystal oxide semiconductor. Examples of the non-single-crystal oxide semiconductors include a CAAC-OS (C-Axis-Aligned Crystalline Oxide Semiconductor), a polycrystalline oxide semiconductor, an nc-OS (nanocrystalline oxide semiconductor), an amorphous-like oxide semiconductor (a-like OS), an amorphous oxide semiconductor, and the like. Among non-single-crystal structures, an amorphous structure has the highest density of defect states, whereas the CAAC-OS has the lowest density of defect states.
(193) An oxide semiconductor film having an amorphous structure has disordered atomic arrangement and no crystalline component, for example. Alternatively, an oxide film having an amorphous structure has, for example, a completely amorphous structure and no crystal part.
(194) Note that the semiconductor layer may be a mixed film including two or more of a region having an amorphous structure, a region having a microcrystalline structure, a region having a polycrystalline structure, a CAAC-OS region, and a region having a single crystal structure. The mixed film has, for example, a single-layer structure or a stacked-layer structure including two or more of the above regions in some cases.
(195) In addition, a CAC-OS (Cloud-Aligned Composite oxide semiconductor) may be used for a semiconductor layer of a transistor disclosed in one embodiment of the present invention.
(196) Note that the non-single-crystal oxide semiconductor or CAC-OS can be suitably used for a semiconductor layer of a transistor disclosed in one embodiment of the present invention. In addition, as the non-single-crystal oxide semiconductor, the nc-OS or the CAAC-OS can be suitably used.
(197) Note that in one embodiment of the present invention, a CAC-OS is preferably used for a semiconductor layer of a transistor. The use of the CAC-OS allows the transistor to have high

electrical characteristics or high reliability.
(198) Note that the semiconductor layer may be a mixed film including two or more kinds of a region of a CAAC-OS, a region of a polycrystalline oxide semiconductor, a region of an nc-OS, a region of an amorphous-like oxide semiconductor, and a region of an amorphous oxide semiconductor. The mixed film has, for example, a single-layer structure or a stacked-layer structure including two or more kinds of the above regions in some cases.
(199) <Composition of CAC-OS>
(200) The composition of a CAC (Cloud-Aligned Composite)-OS that can be used in a transistor disclosed in one embodiment of the present invention is described below.
(201) The CAC-OS is, for example, a composition of a material in which elements that constitute a metal oxide are unevenly distributed to have a size of greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 1 nm and less than or equal to 2 nm, or a similar size. Note that in the following description, a state in which one or more metal elements are unevenly distributed and regions including the metal element(s) are mixed to have a size of greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 1 nm and less than or equal to 2 nm, or a similar size in a metal oxide is referred to as a mosaic pattern or a patch-like pattern.
(202) Note that the metal oxide preferably contains at least indium. In particular, indium and zinc are preferably contained. Moreover, in addition to these, one kind or a plurality of kinds selected from aluminum, gallium, yttrium, copper, vanadium, beryllium, boron, silicon, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like may be contained.
(203) For example, a CAC-OS in an In—Ga—Zn oxide (an In—Ga—Zn oxide in the CAC-OS may be particularly referred to as CAC-IGZO) has a composition in which materials are separated into indium oxide (hereinafter referred to as InO.sub.X1 (X1 is a real number greater than 0)) or indium zinc oxide (hereinafter referred to as In.sub.X2Zn.sub.Y2O.sub.Z2 (each of X2, Y2, and Z2 is a real number greater than 0)) and gallium oxide (hereinafter referred to as GaO.sub.X3 (X3 is a real number greater than 0)), gallium zinc oxide (hereinafter referred to as Ga.sub.X4Zn.sub.Y4O.sub.Z4 (each of X4, Y4, and Z4 is a real number greater than 0)), or the like so that a mosaic pattern is formed, and mosaic-like InO.sub.X1 or In.sub.X2Zn.sub.Y2O.sub.Z2 is evenly distributed in the film (this composition is hereinafter also referred to as a cloud-like composition).
(204) That is, the CAC-OS is a composite metal oxide having a composition in which a region where GaO.sub.X3 is a main component and a region where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component are mixed. Note that in this specification, for example, when the atomic ratio of In to an element M in a first region is larger than the atomic ratio of In to the element M in a second region, the first region is regarded as having a higher In concentration than the second region.
(205) Note that IGZO is a commonly known name and sometimes refers to one compound formed of In, Ga, Zn, and O. A typical example is a crystalline compound represented by InGaO.sub.3(ZnO).sub.m1 (m1 is a natural number) or In.sub.(1+x0)Ga.sub.(1−x0)O.sub.3(ZnO).sub.m0 (−1≤x0≤1; m0 is a given number).
(206) The crystalline compound has a single crystal structure, a polycrystalline structure, or a CAAC structure. Note that the CAAC structure is a crystal structure in which a plurality of IGZO nanocrystals have c-axis alignment and are connected in an a-b plane without alignment.
(207) Meanwhile, the CAC-OS relates to the material composition of a metal oxide. In the material composition of a CAC-OS containing In, Ga, Zn, and O, some regions that contain Ga as a main component and are observed as nanoparticles and some regions that contain In as a main component and are observed as nanoparticles are each randomly dispersed in a mosaic pattern. Therefore, the crystal structure is a secondary element for the CAC-OS.

(208) Note that the CAC-OS is regarded as not including a stacked-layer structure of two or more kinds of films with different compositions. For example, a two-layer structure of a film containing In as a main component and a film containing Ga as a main component is not included.
(209) Note that a clear boundary between the region where GaO.sub.X3 is a main component and the region where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component cannot be observed in some cases.
(210) Note that in the case where one kind or a plurality of kinds selected from aluminum, yttrium, copper, vanadium, beryllium, boron, silicon, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like are contained instead of gallium, the CAC-OS refers to a composition in which some regions that contain the metal element(s) as a main component and are observed as nanoparticles and some regions that contain In as a main component and are observed as nanoparticles are each randomly dispersed in a mosaic pattern.
(211) The CAC-OS can be formed by a sputtering method under a condition where a substrate is not heated intentionally, for example. In addition, in the case of forming the CAC-OS by a sputtering method, one or more selected from an inert gas (typically, argon), an oxygen gas, and a nitrogen gas may be used as a deposition gas. Furthermore, the ratio of the flow rate of an oxygen gas to the total flow rate of the deposition gas at the time of deposition is preferably as low as possible, and for example, the ratio of the flow rate of the oxygen gas is preferably higher than or equal to 0% and lower than 30%, further preferably higher than or equal to 0% and lower than or equal to 10%.
(212) The CAC-OS is characterized in that no clear peak is observed at the time of measurement using θ/2θ scan by an Out-of-plane method, which is one of the X-ray diffraction (XRD) measurement methods. That is, it is found from X-ray diffraction measurement that no alignment in an a-b plane direction and a c-axis direction is observed in a measured region.
(213) In addition, in an electron diffraction pattern of the CAC-OS that is obtained by irradiation with an electron beam with a probe diameter of 1 nm (also referred to as a nanobeam electron beam), a ring-like high-luminance region (ring region) and a plurality of bright spots in the ring region are observed. It is therefore found from the electron diffraction pattern that the crystal structure of the CAC-OS includes an nc (nano-crystal) structure with no alignment in a plan-view direction and a cross-sectional direction.
(214) Moreover, for example, it can be confirmed by EDX mapping obtained using energy dispersive X-ray spectroscopy (EDX) that the CAC-OS in the In—Ga—Zn oxide has a composition in which regions where GaO.sub.X3 is a main component and regions where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component are unevenly distributed and mixed.
(215) The CAC-OS has a composition different from that of an IGZO compound in which metal elements are evenly distributed, and has characteristics different from those of the IGZO compound. That is, the CAC-OS has a composition in which regions where GaO.sub.X3 or the like is a main component and regions where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component are phase-separated from each other, and the regions including the respective elements as the main components form a mosaic pattern.
(216) Here, a region where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component is a region whose conductivity is higher than that of a region where GaO.sub.X3 or the like is a main component. In other words, when carriers flow through regions where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component, the conductivity of a metal oxide is exhibited. Accordingly, when the regions where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component are distributed like a cloud in a metal oxide, high field-effect mobility (μ) can be achieved.
(217) In contrast, a region where GaO.sub.X3 or the like is a main component is a region whose

insulating property is higher than that of a region where In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 is a main component. In other words, when regions where GaO.sub.X3 or the like is a main component are distributed in a metal oxide, leakage current can be suppressed and favorable switching operation can be achieved.
(218) Accordingly, when the CAC-OS is used for a semiconductor element, the insulating property derived from GaO.sub.X3 or the like and the conductivity derived from In.sub.X2Zn.sub.Y2O.sub.Z2 or InO.sub.X1 complement each other, so that high on-state current (Ion) and high field-effect mobility (μ) can be achieved.
(219) In addition, a semiconductor element using the CAC-OS has high reliability. Thus, the CAC-OS is suitable for a variety of semiconductor devices typified by a display.
(220) In addition, since a transistor including the CAC-OS in a semiconductor layer has high field-effect mobility and high drive capability, the use of the transistor in a driver circuit, a typical example of which is a scan line driver circuit that generates a gate signal, can provide a display device with a narrow bezel width (also referred to a narrow bezel). Furthermore, with the use of the transistor in a signal line driver circuit that is included in a display device (particularly in a demultiplexer connected to an output terminal of a shift register included in a signal line driver circuit), a display device to which a small number of wirings are connected can be provided.
(221) Furthermore, unlike a transistor including low-temperature polysilicon, the transistor including the CAC-OS in the semiconductor layer does not need a laser crystallization step. Thus, the manufacturing cost of a display device can be reduced even when the display device is formed using a large area substrate. In addition, the transistor including the CAC-OS in the semiconductor layer is preferably used for a driver circuit and a display portion in a large display device having high resolution such as ultra-high definition ("4K resolution," "4K2K," and "4K") or super high definition ("8K resolution," "8K4K," and "8K") because writing can be performed in a short time and display defects can be reduced.
(222) Alternatively, silicon may be used for a semiconductor in which a channel of a transistor is formed. Although amorphous silicon may be used as silicon, silicon having crystallinity is particularly preferably used. For example, microcrystalline silicon, polycrystalline silicon, single crystal silicon, or the like is preferably used. In particular, polycrystalline silicon can be formed at a temperature lower than that for single crystal silicon and has higher field-effect mobility and higher reliability than amorphous silicon.
(223) <Conductive Layer>
(224) Examples of materials that can be used for conductive layers of a variety of wirings and electrodes and the like included in the display device in addition to a gate, a source, and a drain of a transistor include metals such as aluminum, titanium, chromium, nickel, copper, yttrium, zirconium, molybdenum, silver, tantalum, and tungsten and an alloy containing such a metal as its main component. Alternatively, a single layer or a stacked-layer structure including a film containing these materials can be used. For example, a single-layer structure of an aluminum film containing silicon, a two-layer structure in which an aluminum film is stacked over a titanium film, a two-layer structure in which an aluminum film is stacked over a tungsten film, a two-layer structure in which a copper film is stacked over a copper-magnesium-aluminum alloy film, a two-layer structure in which a copper film is stacked over a titanium film, a two-layer structure in which a copper film is stacked over a tungsten film, a three-layer structure in which an aluminum film or a copper film is stacked over a titanium film or a titanium nitride film and a titanium film or a titanium nitride film is formed thereover, a three-layer structure in which an aluminum film or a copper film is stacked over a molybdenum film or a molybdenum nitride film and a molybdenum film or a molybdenum nitride film is formed thereover, and the like can be given. Note that an oxide such as indium oxide, tin oxide, or zinc oxide may be used. In addition, copper containing manganese is preferably used because controllability of a shape by etching is increased.
(225) <Insulating Layer>

(226) Examples of an insulating material that can be used for each insulating layer include, in addition to a resin such as acrylic or epoxy and a resin having a siloxane bond, an inorganic insulating material such as silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, or aluminum oxide.
(227) In addition, the light-emitting device is preferably provided between a pair of insulating films with low water permeability. In that case, impurities such as water can be inhibited from entering the light-emitting device, and a decrease in the reliability of the device can be inhibited.
(228) Examples of the insulating film with low water permeability include a film containing nitrogen and silicon, such as a silicon nitride film and a silicon nitride oxide film, and a film containing nitrogen and aluminum, such as an aluminum nitride film. Alternatively, a silicon oxide film, a silicon oxynitride film, an aluminum oxide film, or the like may be used.
(229) For example, the moisture vapor transmission rate of the insulating film with low water permeability is lower than or equal to 1×10.sup.−5 [g/(m.sup.2.Math.day)], preferably lower than or equal to 1×10.sup.−6 [g/(m.sup.2.Math.day)], further preferably lower than or equal to 1×10.sup.−7 [g/(m.sup.2.Math.day)], still further preferably lower than or equal to 1×10.sup.−8 [g/(m.sup.2.Math.day)].
(230) The above is the description of the components.
(231) At least part of the structure examples, the drawings corresponding thereto, and the like exemplified in this embodiment can be implemented in combination with the other structure examples, the other drawings, and the like as appropriate.
(232) At least part of this embodiment can be implemented in combination with the other embodiments described in this specification as appropriate.
Embodiment 3
(233) In this embodiment, structure examples of a display device will be described using FIG. **23**A to FIG. **23**C.
(234) The display device illustrated in FIG. **23**A includes a pixel portion **502**, a driver circuit portion **504**, protection circuits **506**, and a terminal portion **507**. Note that a structure in which the protection circuits **506** are not provided may be employed.
(235) The pixel portion **502** includes a plurality of pixel circuits **501** that drive a plurality of display devices arranged in X rows and Y columns (X and Y each independently represent a natural number of 2 or more).
(236) The driver circuit portion **504** includes driver circuits such as a gate driver **504***a* that outputs scan signals to gate lines GL_**1** to GL_X and a source driver **504***b* that supplies data signals to data lines DL_**1** to DL_Y. The gate driver **504***a* includes at least a shift register. In addition, the source driver **504***b* is formed using a plurality of analog switches, for example. Alternatively, the source driver **504***b* may be formed using a shift register or the like.
(237) The terminal portion **507** refers to a portion provided with terminals for inputting power, control signals, image signals, and the like to the display device from external circuits.
(238) The protection circuit **506** is a circuit that, when a potential out of a certain range is applied to a wiring to which the protection circuit **506** is connected, establishes continuity between the wiring and another wiring. The protection circuit **506** illustrated in FIG. **23**A is connected to a variety of wirings such as scan lines GL that are wirings between the gate driver **504***a* and the pixel circuits **501** and data lines DL that are wirings between the source driver **504***b* and the pixel circuits **501**, for example.
(239) In addition, the gate driver **504***a* and the source driver **504***b* may each be provided over the same substrate as the pixel portion **502**, or a substrate over which a gate driver circuit or a source driver circuit is separately formed (e.g., a driver circuit board formed using a single crystal semiconductor film or a polycrystalline semiconductor film) may be mounted on the substrate by COF, TCP (Tape Carrier Package), COG (Chip On Glass), or the like.
(240) In addition, the plurality of pixel circuits **501** illustrated in FIG. **23**A can have a structure

illustrated in FIG. **23**B or FIG. **23**C, for example.
(241) The pixel circuit **501** illustrated in FIG. **23**B includes a liquid crystal device **570**, a transistor **550**, and a capacitor **560**. In addition, the data line DL_n, the scan line GL_m, a potential supply line VL, and the like are connected to the pixel circuit **501**.
(242) The potential of one of a pair of electrodes of the liquid crystal device **570** is set as appropriate in accordance with the specifications of the pixel circuit **501**. The alignment state of the liquid crystal device **570** is set depending on written data. Note that a common potential may be applied to one of the pair of electrodes of the liquid crystal device **570** included in each of the plurality of pixel circuits **501**. Alternatively, a different potential may be applied to one of the pair of electrodes of the liquid crystal device **570** of the pixel circuit **501** in each row.
(243) In addition, the pixel circuit **501** illustrated in FIG. **23**C includes transistors **552** and **554**, a capacitor **562**, and a light-emitting device **572**. Furthermore, the data line DL_n, the scan line GL_m, a potential supply line VL_a, a potential supply line VL_b, and the like are connected to the pixel circuit **501**.
(244) Note that a high power supply potential VDD is applied to one of the potential supply line VL_a and the potential supply line VL_b, and a low power supply potential VSS is applied to the other of the potential supply line VL_a and the potential supply line VL_b. Current flowing through the light-emitting device **572** is controlled in accordance with a potential applied to a gate of the transistor **554**, so that the luminance of light emitted from the light-emitting device **572** is controlled.
(245) At least part of the structure examples, the drawings corresponding thereto, and the like exemplified in this embodiment can be implemented in combination with the other structure examples, the other drawings, and the like as appropriate.
(246) At least part of this embodiment can be implemented in combination with the other embodiments described in this specification as appropriate.
Embodiment 4
(247) A pixel circuit including a memory for correcting gray levels displayed by pixels and a display device including the pixel circuit will be described below.
(248) <Circuit Structure>
(249) FIG. **24**A illustrates a circuit diagram of a pixel circuit **400**. The pixel circuit **400** includes a transistor M**1**, a transistor M**2**, a capacitor C**1**, and a circuit **401**. In addition, a wiring S**1**, a wiring S**2**, a wiring G**1**, and a wiring G**2** are connected to the pixel circuit **400**.
(250) In the transistor M**1**, a gate is connected to the wiring G**1**, one of a source and a drain is connected to the wiring S**1**, and the other of the source and the drain is connected to one electrode of the capacitor C**1**. In the transistor M**2**, a gate is connected to the wiring G**2**, one of a source and a drain is connected to the wiring S**2**, and the other of the source and the drain is connected to the other electrode of the capacitor C**1** and the circuit **401**.
(251) The circuit **401** is a circuit including at least one display device. A variety of devices can be used as the display device, and typically, a light-emitting device such as an organic EL device or an LED device, a liquid crystal device, a MEMS (Micro Electro Mechanical Systems) device, or the like can be employed.
(252) A node connecting the transistor M**1** and the capacitor C**1** is denoted as a node N**1**, and a node connecting the transistor M**2** and the circuit **401** is denoted as a node N**2**.
(253) In the pixel circuit **400**, the potential of the node N**1** can be retained when the transistor M**1** is set in an off state. In addition, the potential of the node N**2** can be retained when the transistor M**2** is set in an off state. Furthermore, when a predetermined potential is written to the node N**1** through the transistor M**1** with the transistor M**2** being in an off state, the potential of the node N**2** can be changed in accordance with displacement in the potential of the node N**1** owing to capacitive coupling through the capacitor C**1**.
(254) Here, the transistor employing an oxide semiconductor, which is illustrated in Embodiment 1,

can be used as one or both of the transistor M**1** and the transistor M**2**. Accordingly, owing to extremely low off-state current, the potentials of the node N**1** and the node N**2** can be retained over a long period. Note that in the case where the period in which the potential of each node is retained is short (specifically, the case where frame frequency is higher than or equal to 30 Hz, for example), a transistor employing a semiconductor such as silicon may be used.

Driving Method Example

(255) Next, an example of a method for operating the pixel circuit **400** is described using FIG. **24**B. FIG. **24**B is a timing chart of the operation of the pixel circuit **400**. Note that for simplification of the description, the influence of various kinds of resistance such as wiring resistance, parasitic capacitance of a transistor, a wiring, or the like, the threshold voltage of the transistor, and the like is not taken into account here.

(256) In the operation shown in FIG. **24**B, one frame period is divided into a period T**1** and a period T**2**. The period T**1** is a period in which a potential is written to the node N**2**, and the period T**2** is a period in which a potential is written to the node N**1**.

(257) In the period T**1**, a potential for setting the transistor in an on state is applied to both the wiring G**1** and the wiring G**2**. In addition, a potential V.sub.ref that is a fixed potential is supplied to the wiring S**1**, and a first data potential V.sub.w is supplied to the wiring S**2**.

(258) The potential V.sub.ref is applied from the wiring S**1** to the node N**1** through the transistor M**1**. In addition, the first data potential V.sub.w is applied from the wiring S**2** to the node N**2** through the transistor M**2**. Accordingly, a potential difference V.sub.w−V.sub.ref is retained in the capacitor C**1**.

(259) Next, in the period T**2**, a potential for setting the transistor M**1** in an on state is applied to the wiring G**1**, and a potential for setting the transistor M**2** in an off state is applied to the wiring G**2**. In addition, a second data potential V.sub.data is supplied to the wiring S**1**. The wiring S**2** may be supplied with a predetermined constant potential or brought into a floating state.

(260) The second data potential V.sub.data is applied from the wiring S**1** to the node N**1** through the transistor M**1**. In that case, capacitive coupling due to the capacitor C**1** changes the potential of the node N**2** in accordance with the second data potential V.sub.data by a potential dV. That is, a potential that is the sum of the first data potential V.sub.w and the potential dV is input to the circuit **401**. Note that although dV is shown as a positive value in FIG. **24**B, dV may be a negative value. That is, the potential V.sub.data may be lower than the potential V.sub.ref.

(261) Here, the potential dV is roughly determined by the capacitance value of the capacitor C**1** and the capacitance value of the circuit **401**. In the case where the capacitance value of the capacitor C**1** is sufficiently larger than the capacitance value of the circuit **401**, the potential dV is a potential close to the second data potential V.sub.data.

(262) As described above, a potential to be supplied to the circuit **401** including the display device can be generated by a combination of two kinds of data signals in the pixel circuit **400**, so that gray levels can be corrected in the pixel circuit **400**.

(263) In addition, in the pixel circuit **400**, it is also possible to generate a potential exceeding the maximum potential that can be supplied to the wiring S**1** and the wiring S**2**. For example, in the case where a light-emitting device is used, high-dynamic range (HDR) display or the like can be performed. Furthermore, in the case where a liquid crystal device is used, overdriving or the like can be achieved.

Application Example

(264) [Example Using Liquid Crystal Device]

(265) A pixel circuit **400**LC illustrated in FIG. **24**C includes a circuit **401**LC. The circuit **401**LC includes a liquid crystal device LC and a capacitor C**2**.

(266) In the liquid crystal device LC, one electrode is connected to the node N**2** and one electrode of the capacitor C**2**, and the other electrode is connected to a wiring supplied with a potential V.sub.com2. The other electrode of the capacitor C**2** is connected to a wiring supplied with a

potential V.sub.com1.
(267) The capacitor C**2** functions as a storage capacitor. Note that the capacitor C**2** can be omitted when not needed.
(268) In the pixel circuit **400**LC, high voltage can be supplied to the liquid crystal device LC; thus, high-speed display can be performed by overdriving or a liquid crystal material with high drive voltage can be employed, for example. In addition, gray levels can also be corrected in accordance with operating temperature, the degradation state of the liquid crystal device LC, or the like by supply of a correction signal to the wiring S**1** or the wiring S**2**.
(269) [Example Using Light-Emitting Device]
(270) A pixel circuit **400**EL illustrated in FIG. **24**D includes a circuit **401**EL. The circuit **401**EL includes a light-emitting device EL, a transistor M**3**, and the capacitor C**2**.
(271) In the transistor M**3**, a gate is connected to the node N**2** and one electrode of the capacitor C**2**, one of a source and a drain is connected to a wiring supplied with a potential V.sub.H, and the other of the source and the drain is connected to one electrode of the light-emitting device EL. The other electrode of the capacitor C**2** is connected to a wiring supplied with a potential V.sub.com. The other electrode of the light-emitting device EL is connected to a wiring supplied with a potential V.sub.L.
(272) The transistor M**3** has a function of controlling current to be supplied to the light-emitting device EL. The capacitor C**2** functions as a storage capacitor. The capacitor C**2** can be omitted when not needed.
(273) Note that although a structure in which the anode side of the light-emitting device EL is connected to the transistor M**3** is described here, the transistor M**3** may be connected to the cathode side. In that case, the values of the potential V.sub.H and the potential V.sub.L can be changed as appropriate.
(274) In the pixel circuit **400**EL, a large amount of current can flow through the light-emitting device EL when a high potential is applied to the gate of the transistor M**3**, which enables HDR display or the like, for example. Moreover, a variation in electrical characteristics of the transistor M**3** and the light-emitting device EL can also be corrected by supply of a correction signal to the wiring S**1** or the wiring S**2**.
(275) Note that the structure is not limited to the circuits illustrated in FIG. **24**C and FIG. **24**D, and a structure to which a transistor, a capacitor, or the like is further added may be employed.
(276) At least part of this embodiment can be implemented in combination with the other embodiments described in this specification as appropriate.
Embodiment 5
(277) In this embodiment, structure examples of a pixel of a display panel according to one embodiment of the present invention will be described below.
(278) Structure examples of a pixel **300** are illustrated in FIG. **25**A to FIG. **25**E.
(279) The pixel **300** includes a plurality of pixels **301**. The plurality of pixels **301** each function as a subpixel. One pixel **300** is formed of the plurality of pixels **301** exhibiting different colors, and thus full-color display can be achieved in a display portion.
(280) The pixels **300** illustrated in FIG. **25**A and FIG. **25**B each include three subpixels. The combination of colors exhibited by the pixels **301** included in the pixel **300** illustrated in FIG. **25**A is red (R), green (G), and blue (B). The combination of colors exhibited by the pixels **301** included in the pixel **300** illustrated in FIG. **25**B is cyan (C), magenta (M), and yellow (Y).
(281) The pixels **300** illustrated in FIG. **25**C to FIG. **25**E each include four subpixels. The combination of colors exhibited by the pixels **301** included in the pixel **300** illustrated in FIG. **25**C is red (R), green (G), blue (B), and white (W). The use of the subpixel that exhibits white can increase the luminance of the display portion. The combination of colors exhibited by the pixels **301** included in the pixel **300** illustrated in FIG. **25**D is red (R), green (G), blue (B), and yellow (Y). The combination of colors exhibited by the pixels **301** included in the pixel **300** illustrated in

FIG. **25**E is cyan (C), magenta (M), yellow (Y), and white (W).

(282) When subpixels that exhibit red, green, blue, cyan, magenta, yellow, and the like are combined as appropriate with more subpixels functioning as one pixel, the reproducibility of halftones can be increased. Thus, display quality can be increased.

(283) In addition, the display device according to one embodiment of the present invention can reproduce the color gamut of various standards. For example, the display device according to one embodiment of the present invention can reproduce the color gamut of the PAL (Phase Alternating Line) standard and the NTSC (National Television System Committee) standard used for TV broadcasting; the sRGB (standard RGB) standard and the Adobe RGB standard widely used for display devices used in electronic devices such as personal computers, digital cameras, and printers; the ITU-R BT.709 (International Telecommunication Union Radiocommunication Sector Broadcasting Service (Television) 709) standard used for HDTV (High Definition Television, also referred to Hi-Vision); the DCI-P3 (Digital Cinema Initiatives P3) standard used for digital cinema projection; the ITU-R BT.2020 (REC.2020 (Recommendation 2020)) standard used for UHDTV (Ultra High Definition Television, also referred to as Super Hi-Vision); and the like.

(284) In addition, by arranging the pixels **300** in a matrix of 1920×1080, a display device that can perform full-color display with a resolution of what is called full high definition (also referred to as "2K resolution," "2K1K," "2K," or the like) can be achieved. Alternatively, for example, by arranging the pixels **300** in a matrix of 3840×2160, a display device that can perform full-color display with a resolution of what is called ultra high definition (also referred to as "4K resolution," "4K2K," "4K," or the like) can be achieved. Alternatively, for example, by arranging the pixels **300** in a matrix of 7680×4320, a display device that can perform full-color display with a resolution of what is called super high definition (also referred to as "8K resolution," "8K4K," "8K," or the like) can be achieved. By increasing the number of pixels **300**, a display device that can perform full-color display with 16K or 32K resolution can also be achieved.

(285) At least part of this embodiment can be implemented in combination with the other embodiments described in this specification as appropriate.

REFERENCE NUMERALS

(286) **10**: hinge, **11***a*: blade, **11***b*: blade, **12***a*: shaft, **12***b*: shaft, **13***a*: shaft, **13***b*: shaft, **14**: spindle, **15**: spindle, **16**: pin, **17**: stopper, **19**: jig, **20**: hinge, **21**: columnar body, **22**: slotted hole, **23**: stopper, **24**: backing plate, **25**: hole portion, **26**: hollow portion, **27**: spring, **28**: ball, **29**: lock part, **30**: notch portion, **31**: region, **41***a*: shaft, **41***b*: shaft, **42**: shaft, **45**: spindle, **46**: pin, **100**A: support, **100**B: support, **101***a*: hinge portion, **101***b*: hinge portion, **102***a*: housing, **102***b*: housing, **102***c*: housing, **103**: display panel, **103***a*: region, **103***b*: region, **103***c*: region, **104***a*: curved surface, **104***b*: curved surface, **115**: columnar body, **116***a*: gear, **116***b*: gear, **120**: sensor, **130**: image, **131**: keyboard, **132**: icon, **135***a*: input/output unit, **135***b*: input/output unit, **136***a*: camera, **136***b*: camera, **137**: sensor, **138**: display panel, **139**: display panel, **140**: solar battery, **141**: thin-film solar battery, **145**: external interface, **146**: transceiving unit, **147**: speaker, **148**: camera, **149**: microphone, **150**: stylus, **200**: display device, **210**: display device, **300**: pixel, **301**: pixel, **400**: pixel circuit, **400**EL: pixel circuit, **400**LC: pixel circuit, **401**: circuit, **401**EL: circuit, **401**LC: circuit, **501**: pixel circuit, **502**: pixel portion, **504**: driver circuit portion, **504***a*: gate driver, **504***b*: source driver, **506**: protection circuit, **507**: terminal portion, **550**: transistor, **552**: transistor, **554**: transistor, **560**: capacitor, **562**: capacitor, **570**: liquid crystal device, **572**: light-emitting device, **700**: display panel, **700**A: display panel, **702**: pixel portion, **704**: source driver circuit portion, **706**: gate driver circuit portion, **708**: FPC terminal portion, **710**: wiring, **716**: FPC, **717**: IC, **730**: insulating layer, **732**: sealing layer, **736**: coloring layer, **738**: light-blocking layer, **740**: support substrate, **741**: protective layer, **741***a*: insulating layer, **741***b*: insulating layer, **741***c*: insulating layer, **742**: adhesive layer, **743**: resin layer, **744**: insulating layer, **745**: support substrate, **746**: insulating layer, **747**: adhesive layer, **749**: protective layer, **750**: transistor, **752**: transistor, **760**: wiring, **761**: conductive layer, **770**: insulating layer, **772**: conductive

layer, **780**: anisotropic conductive film, **782**: light-emitting device, **786**: EL layer, **788**: conductive layer, and **790**: capacitor.

Claims

1. A flexible component support comprising: a first housing; a second housing; a first joint; and a second joint, wherein the first housing and the second housing are coupled to each other through the first joint and the second joint, wherein the first joint and the second joint have an overlap region, wherein the first joint and the second joint each include a movable portion, wherein the first joint includes a plurality of first columnar bodies, wherein the plurality of first columnar bodies are coupled to each other so that first surfaces of the plurality of first columnar bodies form a continuous surface, wherein the second joint is a hinge, wherein a maximum opening angle of the hinge is approximately 180°, and wherein, when the opening angle is 180°, second surfaces of the plurality of first columnar bodies opposite of the first surfaces of the plurality of first columnar bodies face a first surface of the second joint.
2. The flexible component support according to claim 1, wherein a second columnar body including a major axis substantially perpendicular to the first surface of one of the plurality of first columnar bodies is fixed to the one of the plurality of first columnar bodies and the hinge, wherein the hinge includes a notch portion, and wherein when the opening angle of the hinge is approximately 180°, part of the second columnar body is positioned in the notch portion.
3. The flexible component support according to claim 1, wherein the hinge comprises a first blade and a second blade, wherein a first end portion of a second columnar body including a major axis substantially perpendicular to the first surface of one of the plurality of first columnar bodies is fixed to the one of the plurality of first columnar bodies, wherein the hinge includes a plurality of shafts, wherein the first blade is fixed to one of the plurality of shafts, wherein the second blade is not fixed to the one of the plurality of shafts, and wherein a second end portion in the major axis direction of the second columnar body is fixed to the one of the plurality of shafts.
4. A display device comprising: the flexible component support according to claim 1, and a flexible display panel.
5. The display device according to claim 4, wherein the flexible display panel includes a light-emitting device.
6. A flexible component support comprising: a first housing; a second housing; a first joint; and a second joint, wherein the first housing and the second housing are coupled to each other through the first joint and the second joint, wherein the first joint and the second joint have an overlap region, wherein the first joint and the second joint each include a movable portion, wherein the first joint includes a plurality of first columnar bodies, wherein the plurality of first columnar bodies are coupled to each other so that first surfaces of the plurality of first columnar bodies and first surfaces of the first housing and the second housing form a continuous surface, wherein the second joint is a hinge comprising a first blade and a second blade, wherein a maximum opening angle of the hinge is approximately 180°, and wherein the second joint is fixed to a second surface of the first housing opposite of the first surface of the first housing.
7. A flexible component support comprising: a first housing; a second housing; a first joint; and a second joint, wherein the first housing and the second housing are connected to each other through the first joint and the second joint, wherein the first joint and the second joint have an overlap region, wherein the first joint and the second joint each include a movable portion, wherein the first joint includes a plurality of first columnar bodies, wherein the plurality of first columnar bodies are coupled to each other so that first surfaces of the plurality of first columnar bodies form a continuous surface, wherein the second joint is a hinge, and wherein, in an opened state of the flexible component support, the second joint comprises a region overlapping with the first housing and not overlapping with the first joint.

8. The flexible component support according to claim 7, wherein the hinge comprises a first blade and a second blade, wherein a second columnar body is fixed to one of the plurality of first columnar bodies and the hinge, wherein the hinge includes a notch portion, and wherein when a first surface of the first housing and a first surface of the second housing are aligned with each other, part of the second columnar body is positioned in the notch portion.

9. The flexible component support according to claim 7, wherein the hinge comprises a first blade and a second blade, wherein a first end portion of a second columnar body is fixed to one of the plurality of first columnar bodies, wherein the hinge includes a plurality of shafts, wherein the first blade is fixed to one of the plurality of shafts, wherein the second blade is not fixed to the one of the plurality of shafts, and wherein a second end portion of the second columnar body is fixed to the one of the plurality of shafts.